40

Introduction to Quantum Physics

CHAPTER OUTLINE

40.1 Blackbody Radiation and Planck's Hypothesis

40.2 The Photoelectric Effect

40.3 The Compton Effect

40.4 The Nature of Electromagnetic Waves

40.5 The Wave Properties of Particles

40.6 A New Model: The Quantum Particle

40.7 The Double-Slit Experiment Revisited

40.8 The Uncertainty Principle

* An asterisk indicates a question or problem item new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ40.1 The ranking is d > a = e > b > c. The wavelength is described by $\lambda = h/p$ in all cases. For photons, the momentum is given by $p = E/c$, so (a) is the same as (e), and (d) has a wavelength ten times larger. For the particles with mass, $pc = (E^2 - m^2c^4)^{1/2} = ([K + mc^2]^2 - m^2c^4)^{1/2} = (K^2 + 2Kmc^2)^{1/2}$. Thus a particle with larger mass has more momentum for the same kinetic energy, and a shorter wavelength.

OQ40.2 Answer (a). The x-ray photon transfers some of its energy to the electron. Thus, its frequency must decrease.

OQ40.3 Answer (b). In Compton scattering, a photon of energy $E = hf = hc/\lambda$ is scattered from an electron at rest. The scattering sets the electron into motion: the electron gains kinetic energy, so the photon loses energy. Because the photon has less energy, its frequency is smaller than E/h and its wavelength is larger than hc/E.

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OQ40.4 **(i)** Answer (d). Because $P = IV$, the power input to the filament has increased by $8 \times 2 = 16$ times. The filament radiates this greater power according to Stefan's law, so its absolute temperature is higher by the fourth root of 16: it is two times higher.

(ii) Answer (d). By Wien's displacement law, the wavelength emitted with the highest intensity is inversely proportional the temperature: the temperature is twice as large, so the wavelength is half as large.

OQ40.5 Answer (a) and (c). One form of Heisenberg's uncertainty relation is $\Delta x \, \Delta p_x \geq \hbar/2\pi$, which says that one cannot determine both the position and momentum of a particle with arbitrary accuracy. Another form of this relation is $\Delta E \, \Delta t \geq \hbar/2\pi$, which sets a limit on how accurately the energy can be determined in a finite time interval.

OQ40.6 Answer: (a). The stopping potential is 1.00 V, so the maximum kinetic energy is 1.00 eV. From Equation 40.9,

$$K_{\max} = hf - \phi = hc/\lambda - \phi$$

$$\lambda = \frac{hc}{\phi + K_{\max}} = \frac{1240 \text{ eV} \cdot \text{nm}}{(2.50 \text{ eV} + 1.00 \text{ eV})} = 354 \text{ nm}$$

OQ40.7 Answer (c). UV light has the highest frequency of the three, and hence each photon delivers more energy to a skin cell. This explains why you can become sunburned on a cloudy day: clouds block visible light and infrared, but not much ultraviolet. You usually do not become sunburned through window glass, even though you can see the visible light from the Sun coming through the window, because the glass absorbs much of the ultraviolet and reemits it as infrared.

OQ40.8 Answer (d). Electron diffraction by crystals, first detected by the Davisson-Germer experiment in 1927, confirmed de Broglie's hypothesis and, of the listed choices, most clearly demonstrates the wave nature of electrons.

OQ40.9 Answer (c). We obtain the momentum of the electron from

$$K = \frac{1}{2}mu^2 = \frac{p^2}{2m} = e\Delta V \qquad \rightarrow \qquad p = \sqrt{2me\Delta V}$$

The de Broglie wavelength is then

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2me\Delta V}}$$

$$= \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{\sqrt{2\left(9.11 \times 10^{-31} \text{ kg}\right)\left(1.60 \times 10^{-19} \text{ C}\right)(50.0 \text{ V})}}$$

$$= 1.74 \times 10^{-10} \text{ m} = 0.174 \text{ nm}$$

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OQ40.10 The ranking is: electron, proton, helium nucleus. The comparative masses of the particles of interest are $m_p \approx 1\,840 m_e$ and $m_{He} \approx 4m_p$. Assuming each particle is classical, its wavelength is inversely proportional to its mass: $\lambda = h/p = h/mv$.

OQ40.11 **(i)** (a) and (c). Electrons and protons possess mass, therefore they have rest energy $E_R = mc^2$. Photons do not have rest energy—they are never at rest.

(ii) (a) and (c). The electron and the proton have charges $-e$ and $+e$, respectively; the photon has no charge.

(iii) (a), (b), and (c). The electron and proton carry energy $E = \sqrt{p^2c^2 + \left(mc^2\right)^2} = K + mc^2$; the photon carries energy $E = hf$.

(iv) (a), (b), and (c). The electron and proton carry momentum $p = \gamma mu$, the photon carries momentum $p = E/c$, where E is its energy.

(v) Answer (b). Because it is light.

(vi) (a), (b), and (c). Each has the same de Broglie wavelength $\lambda = h/p$.

OQ40.12 Answer (a). If we set $K = \frac{1}{2}mu^2 = \frac{p^2}{2m} = e\Delta V$, which is the same for both particles, then we see that the momentum is $p = \sqrt{2me\Delta V}$, so the electron has the smaller momentum and therefore the longer wavelength $\left(\lambda = \frac{h}{p} = \frac{h}{\sqrt{2me\Delta V}}\right)$.

OQ40.13 Answer (b). Diffraction, polarization, interference, and refraction are all processes associated with waves. However, to understand the photoelectric effect, we must think of the energy transmitted as light coming in discrete packets, or quanta, called photons. Thus, the photoelectric effect most clearly demonstrates the particle nature of light.

OQ40.14 Answer (c). For the same uncertainty in speed, the particle with the smaller mass has the smaller uncertainty in momentum, $\Delta p_x = m\Delta v_x$, thus greater uncertainty in its position: $\Delta x \geq \frac{\hbar}{2\pi\Delta p_x} = \frac{\hbar}{2\pi m\Delta v_x}$. The mass of the electron is smaller than that of the proton, thus its minimum possible uncertainty in position is greater than that of the proton.

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ANSWERS TO CONCEPTUAL QUESTIONS

CQ40.1 In general, a turn of wire receives energy by two energy transfer mechanisms: (1) electrical transmission and (2) absorption of electromagnetic radiation from neighboring turns. Each turn of wire emits radiation similar to blackbody radiation. For most turns, the electromagnetic radiation absorbed comes from two neighbors. The turns on the end, however, have only one neighbor so they receive less energy input by electromagnetic radiation than the others. As a result, they operate at a lower temperature and do not glow as brightly.

CQ40.2 The Compton effect describes the *scattering* of photons from electrons, while the photoelectric effect predicts the ejection of electrons due to the *absorption* of photons by a material.

CQ40.3 Any object of macroscopic size—including a grain of dust—has an undetectably small wavelength, so any diffraction effects it might exhibit are very small, effectively undetectable. Recall historically how the diffraction of sound waves was at one time well known, but the diffraction of light was not.

CQ40.4 No. The second metal may have a larger work function than the first, in which case the incident photons may not have enough energy to eject photoelectrons.

CQ40.5 The stopping potential measures the kinetic energy of the most energetic photoelectrons. Each of them has gotten its energy from a single photon. According to Planck's $E = hf$, the photon energy depends on the frequency of the light. The intensity controls only the number of photons reaching a unit area in a unit time.

CQ40.6 Wave theory predicts that the photoelectric effect should occur at any frequency, provided the light intensity is high enough, or provided that the light shines on the surface for a sufficient time interval so that enough energy is delivered to the surface to eject electrons. However, as seen in the photoelectric experiments, the light must have a sufficiently high frequency for the effect to occur, and that electrons are either ejected almost immediately (less than 10^9 seconds after the surface is illuminated) or not at all, regardless of the intensity.

CQ40.7 Ultraviolet light has shorter wavelength and higher photon energy than any wavelength of visible light.

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CQ40.8 Our eyes are not able to detect all frequencies of electromagnetic waves. For example, all objects that are above 0 K in temperature emit electromagnetic radiation in the infrared region. This describes *everything* in a dark room. We are only able to see objects that emit or reflect electromagnetic radiation in the visible portion of the spectrum.

CQ40.9 An electron has both classical-wave and classical-particle characteristics. In single- and double-slit diffraction and interference experiments, electrons behave like classical waves. An electron has mass and charge. It carries kinetic energy and momentum in parcels of definite size, as classical particles do. At the same time it has a particular wavelength and frequency. Since an electron displays characteristics of both classical waves and classical particles, it is neither a classical wave nor a classical particle. It is customary to call it a *quantum particle*, but another invented term, such as "wavicle," could serve equally well.

CQ40.10 A photon can interact with the photographic film at only one point. A few photons would only give a few dots of exposure, apparently randomly scattered.

CQ40.11 The wavelength of violet light is on the order of $\frac{1}{2}\ \mu\text{m},$ while the de Broglie wavelength of an electron can be 4 orders of magnitude smaller. The resolution is better (recall Rayleigh's criterion) because the diffraction effects are smaller.

CQ40.12 Light has both classical-wave and classical-particle characteristics. In single- and double-slit experiments light behaves like a wave. In the photoelectric effect light behaves like a particle. Light may be characterized as an electromagnetic wave with a particular wavelength or frequency, yet at the same time light may be characterized as a stream of photons, each carrying a discrete energy, hf. Since light displays *both* wave and particle characteristics, perhaps it would be fair to call light a "wavicle." It is customary to call a photon a *quantum particle*, different from a classical particle.

CQ40.13 Comparing Equation 40.9 with the slope-intercept form of the equation for a straight line, $y = mx + b$, we see

(a) that the slope in Figure 40.11 in the text is Planck's constant h and

(b) that the y intercept is $-\phi$, the negative of the work function.

(c) If a different metal were used, the slope would remain the same but the work function would be different. Thus, data for different metals appear as parallel lines on the graph.

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CQ40.14 The discovery of electron diffraction by Davisson and Germer was a fundamental advance in our understanding of the motion of material particles. Newton's laws fail to properly describe the motion of an object with small mass. It moves as a wave, not as a classical particle. Proceeding from this recognition, the development of quantum mechanics made possible describing the motion of electrons in atoms; understanding molecular structure and the behavior of matter at the atomic scale, including electronics, photonics, and engineered materials; accounting for the motion of nucleons in nuclei; and studying elementary particles.

CQ40.15 The spacing between repeating structures on the surface of the feathers or scales is on the order of 1/2 the wavelength of light. An optical microscope would not have the resolution to see such fine detail, while an electron microscope can. The electrons can have much shorter wavelength.

CQ40.16 The *intensity* of electron waves in some small region of space determines the *probability* that an electron will be found in that region.

CQ40.17 The first flaw is that the Rayleigh–Jeans law predicts that the intensity of short wavelength radiation emitted by a black body approaches infinity as the wavelength decreases. This is known as the *ultraviolet catastrophe*. The second flaw is the prediction of much more power output from a black body than is shown experimentally. The intensity of radiation from the black body is given by the area under the red $I(\lambda, T)$ *vs.* λ curve in Figure 40.5 in the text, not by the area under the blue curve.

Planck's Law dealt with both of these issues and brought the theory into agreement with the experimental data by adding an exponential term to the denominator that depends on $1/\lambda$. This keeps both the predicted intensity from approaching infinity as the wavelength decreases and the area under the curve finite.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 40.1 Blackbody Radiation and Planck's Hypothesis

P40.1 The absolute temperature of the heating element is

$$T = 150°\text{C} + 273 = 423\ \text{K}$$

The peak wavelength is, from Equation 40.2,

$$\lambda_{\max} T = 2.898 \times 10^{-3}\ \text{m} \cdot \text{K}$$

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}\ \text{m} \cdot \text{K}}{T} = \frac{2.898 \times 10^{-3}\ \text{m} \cdot \text{K}}{423\ \text{K}} = 6.85 \times 10^{-6}\ \text{m}$$

or $\boxed{6.85\ \mu\text{m, which is in the infrared region of the spectrum.}}$

P40.2 (a) From Equation 40.2,

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}\ \text{m} \cdot \text{K}}{2900\ \text{K}} = \boxed{999\ \text{nm}}$$

(b) The wavelength emitted at the greatest intensity is in the infrared (greater than 700 nm), and according to the graph in Active Figure 40.3, much more energy is radiated at wavelengths longer than $\lambda_{\max}$ than at shorter wavelengths.

P40.3 (a) For lightning,

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}\ \text{m} \cdot \text{K}}{T} \sim \frac{2.898 \times 10^{-3}\ \text{m} \cdot \text{K}}{10^{4}\ \text{K}} \boxed{\sim 10^{-7}\ \text{m}}$$

For the explosion,

$$\lambda_{\max} \sim \frac{2.898 \times 10^{-3}\ \text{m} \cdot \text{K}}{10^{7}\ \text{K}} \boxed{\sim 10^{-10}\ \text{m}}$$

(b) $\boxed{\text{Lightning: ultraviolet; explosion: x-ray and gamma ray}}$

P40.4 (a) The peak radiation occurs at approximately 560 nm wavelength. From Wien's displacement law,

$$T = \frac{0.289\ 8 \times 10^{-2}\ \text{m} \cdot \text{K}}{\lambda_{\max}} = \frac{0.289\ 8 \times 10^{-2}\ \text{m} \cdot \text{K}}{560 \times 10^{-9}\ \text{m}} \approx \boxed{5\ 200\ \text{K}}$$

(b) Clearly, a firefly is not at this temperature, so $\boxed{\text{this is not blackbody radiation}}$.

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P40.5 The energy of a single 500-nm photon is:

$$E_\gamma = hf = \frac{hc}{\lambda} = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}{500\times10^{-9}\ \text{m}}$$
$$= 3.98\times10^{-19}\ \text{J}$$

The energy entering the eye each second

$$E = P\Delta t = IA\Delta t$$
$$= \left(4.00\times10^{-11}\ \text{W/m}^2\right)\left[\frac{\pi}{4}\left(8.50\times10^{-3}\ \text{m}\right)^2\right](1.00\ \text{s})$$
$$= 2.27\times10^{-15}\ \text{J}$$

The number of photons required to yield this energy is

$$n = \frac{E}{E_\gamma} = \frac{2.27\times10^{-15}\ \text{J}}{3.98\times10^{-19}\ \text{J/photon}} = \boxed{5.71\times10^{3}\ \text{photons}}$$

P40.6 (i) Planck's equation is $E = hf$. The photon energies are:

(a) $$E = hf = \left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(620\times10^{12}\ \text{s}^{-1}\right)\left(\frac{1.00\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)$$
$$= \boxed{2.57\ \text{eV}}$$

(b) $$E = hf = \left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.10\times10^{9}\ \text{s}^{-1}\right)\left(\frac{1.00\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)$$
$$= \boxed{1.28\times10^{-5}\ \text{eV}}$$

(c) $$E = hf = \left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(46.0\times10^{6}\ \text{s}^{-1}\right)\left(\frac{1.00\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)$$
$$= \boxed{1.91\times10^{-7}\ \text{eV}}$$

(ii) Wavelengths:

(a) $$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\ \text{m/s}}{620\times10^{12}\ \text{Hz}} = 4.84\times10^{-7}\ \text{m} = \boxed{484\ \text{nm}}$$

(b) $$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\ \text{m/s}}{3.10\times10^{9}\ \text{Hz}} = 9.68\times10^{-2}\ \text{m} = \boxed{9.68\ \text{cm}}$$

(c) $$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\ \text{m/s}}{46.0\times10^{6}\ \text{Hz}} = \boxed{6.52\ \text{m}}$$

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(iii) Part of spectrum:

(a) $\boxed{\text{visible light (blue)}}$

(b) $\boxed{\text{radio wave}}$

(c) $\boxed{\text{radio wave}}$

P40.7 From Wien's displacement law,

(a) $$T = \frac{2.898\times10^{-3}\ \text{m}\cdot\text{K}}{\lambda_{\max}} = \frac{2.898\times10^{-3}\ \text{m}\cdot\text{K}}{970\times10^{-9}\ \text{m}} \approx \boxed{2.99\times10^{3}\ \text{K}}$$

(b) $$T = \frac{2.898\times10^{-3}\ \text{m}\cdot\text{K}}{\lambda_{\max}} = \frac{2.898\times10^{-3}\ \text{m}\cdot\text{K}}{145\times10^{-9}\ \text{m}} \approx \boxed{2.00\times10^{4}\ \text{K}}$$

P40.8 Each photon has an energy

$$E = hf = \left(6.626\times10^{-34}\right)\left(99.7\times10^{6}\right) = 6.61\times10^{-26}\ \text{J}$$

This implies that there are

$$\frac{150\times10^{3}\ \text{J/s}}{6.61\times10^{-26}\ \text{J/photon}} = \boxed{2.27\times10^{30}\ \text{photons/s}}$$

P40.9 From Equation 40.2, Wien's displacement law,

$$T = \frac{2.898\times10^{-3}\ \text{m}\cdot\text{K}}{560\times10^{-9}\ \text{m}} = \boxed{5.18\times10^{3}\ \text{K}}$$

P40.10 (a) From Stefan's law (Equation 40.1), $P = eA\sigma T^4$. If the sun emits as a black body, $e = 1$.

$$T = \left(\frac{P}{eA\sigma}\right)^{1/4}$$

$$= \left[\frac{3.85\times10^{26}\ \text{W}}{1\left[4\pi\left(6.96\times10^{8}\ \text{m}\right)^{2}\right]\left(5.67\times10^{-8}\ \text{W/m}^2\cdot\text{K}^4\right)}\right]^{1/4}$$

$$= \boxed{5.78\times10^{3}\ \text{K}}$$

(b) $$\lambda_{\max} = \frac{2.898\times10^{-3}\ \text{m}\cdot\text{K}}{T} = \frac{2.898\times10^{-3}\ \text{m}\cdot\text{K}}{5.78\times10^{3}\ \text{K}}$$

$$= 5.01\times10^{-7}\ \text{m} = \boxed{501\ \text{nm}}$$

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P40.11 Planck's radiation law, Equation 40.6, gives the intensity-per-wavelength (W/m^2-wavelength). Because the range of the wavelengths is small, we treat the wavelength as the average $\bar{\lambda} = (\lambda_1 + \lambda_2)/2$. Taking E to be the average photon energy and n to be the number of photons emitted each second, we multiply by area and wavelength range to have energy-per-time leaving the hole:

$$\begin{aligned} P &= I(\bar{\lambda}, T)\bar{\lambda} A \\ &= \frac{2\pi hc^2}{\left[(\lambda_1 + \lambda_2)/2\right]^5 \left(e^{2hc/\left[(\lambda_1+\lambda_2)k_B T\right]} - 1 \right)} (\lambda_2 - \lambda_1)\pi (d/2)^2 \\ &= En = nhf \end{aligned}$$

where $$\bar{E} \approx h\bar{f} \approx \frac{hc}{\bar{\lambda}} = \frac{2hc}{\lambda_1 + \lambda_2}$$

Solving for n,

$$n = \frac{P}{E} = \frac{8\pi^2 cd^2 (\lambda_2 - \lambda_1)}{(\lambda_1 + \lambda_2)^4 \left(e^{2hc/\left[(\lambda_1+\lambda_2)k_B T\right]} - 1 \right)}$$

Substituting numerical values and suppressing units,

$$n = \frac{8\pi^2 (3.00\times10^8 \text{ m/s})(0.050\ 0\times10^{-3} \text{ m})^2 (1.00\times10^{-9} \text{ m})}{(1\,001\times10^{-9} \text{ m})^4 \left(e^{\frac{2(6.626\times10^{-34})(3.00\times10^8)}{(1\,001\times10^{-9})(1.38\times10^{-23})(7.50\times10^3)}} - 1 \right)}$$

$$n = \frac{5.90\times10^{16}/\text{s}}{(e^{3.84} - 1)} = \boxed{1.30\times10^{15}/\text{s}}$$

P40.12 (a) From Stefan's law,

$$\begin{aligned} P &= eA\sigma T^4 \\ &= 1(20.0\times10^{-4} \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\cdot\text{K}^4)(5\,000 \text{ K})^4 \\ &= \boxed{7.09\times10^4 \text{ W}} \end{aligned}$$

(b) From Wien's displacement law,

$$\lambda_{\max} T = \lambda_{\max}(5\,000 \text{ K}) = 2.898\times10^{-3} \text{ m}\cdot\text{K} \Rightarrow \lambda_{\max} = \boxed{580 \text{ nm}}$$

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(c) We compute:

$$\frac{hc}{k_B T} = \frac{(6.626\times10^{-34}\ \text{J}\cdot\text{s})(3.00\times10^{8}\ \text{m/s})}{(1.38\times10^{-23}\ \text{J/K})(5\,000\ \text{K})} = 2.88\times10^{-6}\ \text{m}$$

The power per wavelength interval is

$$P(\lambda) = AI(\lambda) = \frac{2\pi hc^2 A}{\lambda^5\left[\exp(hc/\lambda k_B T)-1\right]},$$

and

$$2\pi hc^2 A = 2\pi(6.626\times10^{-34}\ \text{J}\cdot\text{s})(3.00\times10^{8}\ \text{m/s})^2(20.0\times10^{-4}\ \text{m}^2)$$
$$= 7.50\times10^{-19}\ \text{J}\cdot\text{m}^4/\text{s}$$

$$P(580\ \text{nm}) = \frac{7.50\times10^{-19}\ \text{J}\cdot\text{m}^4/\text{s}}{(580\times10^{-9}\ \text{m})^5\left[\exp(2.88\ \mu\text{m}/0.580\ \mu\text{m})-1\right]}$$
$$= \frac{1.15\times10^{13}\ \text{J/m}\cdot\text{s}}{e^{4.973}-1}$$
$$= \boxed{7.99\times10^{10}\ \text{W/m}}$$

(d)–(i) The other values are computed similarly:

	λ	$\frac{hc}{\lambda k_B T}$	$e^{hc/\lambda k_B T}-1$	$\frac{2\pi hc^2 A}{\lambda^5}$	$\mathcal{P}(\lambda)$, W/m
(d)	1.00 nm	2882.6	7.96×10^{1251}	7.50×10^{26}	9.42×10^{-1226}
(e)	5.00 nm	576.5	2.40×10^{250}	2.40×10^{23}	1.00×10^{-227}
(f)	400 nm	7.21	1347	7.32×10^{13}	5.44×10^{10}
(g)	700 nm	4.12	60.4	4.46×10^{12}	7.38×10^{10}
(h)	1.00 mm	0.00288	0.00289	7.50×10^{-4}	0.260
(i)	10.0 cm	2.88×10^{-5}	2.88×10^{-5}	7.50×10^{-14}	2.60×10^{-9}

(j) We approximate the area under the $P(\lambda)$ versus λ curve, between 400 nm and 700 nm, as the product of the average power per wavelength times the range of wavelength:

$$P = P(\bar{\lambda})\Delta\lambda$$
$$= \frac{\left[(5.44+7.38)\times10^{10}\ \text{W/m}\right]}{2}\left[(700-400)\times10^{-9}\ \text{m}\right]$$
$$= 1.92\times10^{4}\ \text{W}\ \boxed{\approx 19\ \text{kW}}$$

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P40.13 (a) The mass of the sphere is

$$m = \rho V = \rho\left(\frac{4}{3}\pi r^3\right) = \left(7.86\times10^3\ \text{kg/m}^3\right)\left[\frac{4}{3}\pi\left(0.020\ 0\ \text{m}\right)^3\right]$$

$$= \boxed{0.263\ \text{kg}}$$

(b) From Stefan's law,

$$P = \sigma AeT^4 = \sigma\left(4\pi r^2\right)eT^4$$

$$P = \left(5.67\times10^{-8}\ \text{W/m}^2\cdot\text{K}^4\right)\left[4\pi\left(0.020\ 0\ \text{m}\right)^2\right](0.860)(293\ \text{K})^4$$

$$= \boxed{1.81\ \text{W}}$$

(c) It emits but does not absorb radiation, so its temperature must drop according to

$$Q = mc\Delta T = mc\left(T_f - T_i\right) \qquad \rightarrow \qquad \frac{dQ}{dt} = mc\frac{dT_f}{dt}$$

$$\frac{dT_f}{dt} = \frac{dQ/dt}{mc} = \frac{-P}{mc}$$

$$= \frac{-1.81\ \text{J/s}}{\left(0.263\ \text{kg}\right)\left(448\ \text{J/kg}\cdot\text{C°}\right)}$$

$$= \boxed{-0.015\ 3\ \text{°C/s} = -0.919\ \text{°C/min}}$$

(d) $\lambda_{\max}T = 2.898\times10^{-3}\ \text{m}\cdot\text{K}$

$$\lambda_{\max} = \frac{2.898\times10^{-3}\ \text{m}\cdot\text{K}}{293\ \text{K}} = 9.89\times10^{-6}\ \text{m} = \boxed{9.89\ \mu\text{m}}\ \text{(infrared)}$$

(e) $$E = hf = \frac{hc}{\lambda} = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.00\times10^8\ \text{m/s}\right)}{9.89\times10^{-6}\ \text{m}} = \boxed{2.01\times10^{-20}\ \text{J}}$$

(f) The energy output each second is carried by photons according to

$$P = \left(\frac{N}{\Delta t}\right)E$$

$$\frac{N}{\Delta t} = \frac{P}{E} = \frac{1.81\ \text{J/s}}{2.01\times10^{-20}\ \text{J/photon}} = \boxed{8.98\times10^{19}\ \text{photon/s}}$$

Matter is coupled to radiation quite strongly, in terms of photon numbers.

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P40.14 Planck's radiation law is

$$I(\lambda, T) = \frac{2\pi hc^2}{\lambda^5\left(e^{hc/\lambda k_B T} - 1\right)}.$$

For long wavelengths, the exponent $hc/\lambda k_B T$ is small. Using the series expansion

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Planck's law reduces to

$$I(\lambda, T) = \frac{2\pi hc^2}{\lambda^5\left[\left(1 + hc/\lambda k_B T + \cdots\right) - 1\right]} \approx \frac{2\pi hc^2}{\lambda^5\left(hc/\lambda k_B T\right)} = \frac{2\pi ck_B T}{\lambda^4}$$

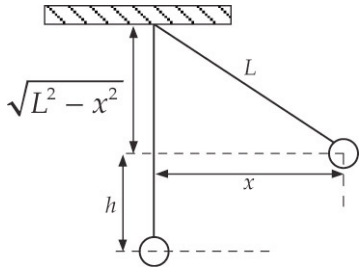
which is the Rayleigh–Jeans law, for very long wavelengths.

P40.15 From the figure, at maximum horizontal displacement x, the bob is at height $h = L - \sqrt{L^2 - x^2}$. Then the pendulum's total energy is

$$E = mgh = mg\left(L - \sqrt{L^2 - x^2}\right)$$

$$E = (1.00\text{ kg})(9.80\text{ m/s}^2)\left(1.00\text{ m} - \sqrt{(1.00\text{ m})^2 - (0.030\ 0\text{ m})^2}\right)$$

$$= 4.41 \times 10^{-3}\text{ J}$$



ANS. FIG. P40.15

The frequency of oscillation is

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{g}{L}} = \frac{1}{2\pi}\sqrt{\frac{9.80\text{ m/s}^2}{1.00\text{ m}}} = 0.498\text{ Hz}$$

The energy is quantized:

$$E = nhf$$

Therefore,

$$n = \frac{E}{hf} = \frac{4.41 \times 10^{-3}\text{ J}}{\left(6.626 \times 10^{-34}\text{ J}\cdot\text{s}\right)\left(0.498\text{ s}^{-1}\right)}$$

$$= \boxed{1.34 \times 10^{31}}$$

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***P40.16** (a) The physical length of the pulse is

$$\ell = vt = \left(3.00\times10^{8}\ \text{m/s}\right)\left(14.0\times10^{-12}\ \text{s}\right) = \boxed{4.20\ \text{mm}}$$

(b) We find the number of photons from

$$E = \frac{hc}{\lambda} = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}{694.3\times10^{-9}\ \text{m}} = 2.86\times10^{-19}\ \text{J}$$

Then,

$$N = \frac{3.00\ \text{J}}{2.86\times10^{-19}\ \text{J}} = \boxed{1.05\times10^{19}\ \text{photons}}$$

(c) The volume of the beam is

$$V = (4.20\ \text{mm})\left[\pi(3.00\ \text{mm})^{2}\right] = 119\ \text{mm}^{3}$$

The number of photons per cubic millimeter is

$$n = \frac{1.05\times10^{19}\ \text{photons}}{119\ \text{mm}^{3}} = \boxed{8.82\times10^{16}\ \text{mm}^{-3}}$$

Section 40.2 The Photoelectric Effect

***P40.17** (a) The cutoff wavelength is given by Equation 40.12:

$$\lambda_c = \frac{hc}{\phi} = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(2.998\times10^{8}\ \text{m/s}\right)}{(4.20\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)} = \boxed{295\ \text{nm}}$$

which corresponds to a frequency of

$$f_c = \frac{c}{\lambda_c} = \frac{2.998\times10^{8}\ \text{m/s}}{295\times10^{-9}\ \text{m}} = \boxed{1.02\times10^{15}\ \text{Hz}}$$

(b) We find the stopping potential from $\frac{hc}{\lambda} = \phi + e\Delta V_S$:

$$\frac{\left(6.626\times10^{-34}\right)\left(2.998\times10^{8}\right)}{180\times10^{-9}} = (4.20\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right) + \left(1.602\times10^{-19}\right)\Delta V_S$$

Therefore, $\boxed{\Delta V_S = 2.69\ \text{V}}$.

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P40.18 (a) At the cutoff wavelength, the energy of the photons is equal to the work function ($K_{max} = 0$):

$$\frac{hc}{\lambda} = \phi \qquad \rightarrow \qquad \lambda = \frac{hc}{\phi} = \frac{1\,240 \text{ nm} \cdot \text{eV}}{4.31 \text{ eV}} = \boxed{288 \text{ nm}}$$

(b) This is the cutoff frequency:

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m/s}}{288 \times 10^{-9} \text{ m}} = \boxed{1.04 \times 10^{15} \text{ Hz}}$$

(c) The maximum kinetic energy is the difference between the energy of the photons and the work function:

$$K_{max} = E - \phi = 5.50 \text{ eV} - 4.31 \text{ eV} = \boxed{1.19 \text{ eV}}$$

P40.19 (a) Einstein's photoelectric effect equation is $K_{max} = hf - \phi$ and the energy required to raise an electron through a 1-V potential is 1 eV, so that

$$K_{max} = e\Delta V_s = 0.376 \text{ eV}$$

The energy of a photon from the mercury lamp is:

$$hf = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})}{546.1 \times 10^{-9} \text{ m}} \left(\frac{1 \text{ eV}}{1.602 \times 10^{-19} \text{ J}} \right)$$

$$= \frac{1\,240 \text{ eV} \cdot \text{nm}}{546.1 \text{ nm}} = 2.27 \text{ eV}$$

Therefore, the work function for this metal is:

$$\phi = hf - K_{max} = 2.27 \text{ eV} - 0.376 \text{ eV} = \boxed{1.89 \text{ eV}}$$

(b) For the yellow light, $\lambda = 587.5$ nm and the photon energy is

$$hf = \frac{hc}{\lambda} = \frac{1\,240 \text{ eV} \cdot \text{nm}}{587.5 \text{ nm}} = 2.11 \text{ eV}$$

Therefore the maximum energy that can be given to an ejected electron is

$$K_{max} = hf - \phi = 2.11 \text{ eV} - 1.89 \text{ eV} = 0.216 \text{ eV}$$

so the stopping voltage is

$$\Delta V_s = \boxed{0.216 \text{ V}}$$

P40.20 (a) The energy of a photon with a wavelength of 400 nm is

$$E = \frac{hc}{\lambda} = \frac{\left(6.63 \times 10^{-34} \text{ J} \cdot \text{s}\right)\left(3.00 \times 10^8 \text{ m/s}\right)}{400 \times 10^{-9} \text{ m}} \left(\frac{1 \text{ eV}}{1.60 \times 10^{-19} \text{ J}} \right)$$

$$= 3.11 \text{ eV}$$

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The energy of a photon with wavelength 400 nm is calculated to be 3.11 eV. Now compare this energy with the given work functions. Of these metals, only lithium shows the photoelectric effect because its work function is less than the energy of the photon.

(b) For lithium,

$$K_{\max} = E - \phi$$

$$= \frac{(6.63\times10^{-34}\ \text{J}\cdot\text{s})(3.00\times10^{8}\ \text{m/s})}{400\times10^{-9}\ \text{m}}\left(\frac{1\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right) - 2.30\ \text{eV}$$

$$= \boxed{0.808\ \text{eV}}$$

P40.21 The maximum kinetic energy of the electrons is

$$K_{\max} = \frac{1}{2}mu_{\max}^2 = \frac{1}{2}(9.11\times10^{-31}\text{kg})(4.60\times10^{5}\text{m/s})^2$$

$$= 9.64\times10^{-20}\ \text{J} = 0.602\ \text{eV}$$

(a) The work function is

$$\phi = E - K_{\max} = \frac{1\ 240\ \text{eV}\cdot\text{nm}}{625\ \text{nm}} - 0.602\ \text{eV} = \boxed{1.38\ \text{eV}}$$

(b) At the cutoff frequency, the energy of the photons equals the work function:

$$E = hf = \phi \quad \rightarrow \quad f = \frac{\phi}{h} = \frac{1.38\ \text{eV}}{6.626\times10^{-34}\ \text{J}\cdot\text{s}}\left(\frac{1.60\times10^{-19}\ \text{J}}{1\ \text{eV}}\right)$$

$$= \boxed{3.34\times10^{14}\ \text{Hz}}$$

P40.22 (a) The energy needed is $E = 1.00\ \text{eV} = 1.60\times10^{-19}\ \text{J}$.

The energy absorbed in time interval Δt is

$$E = P\Delta t = IA\Delta t$$

So,

$$\Delta t = \frac{E}{IA} = \frac{1.60\times10^{-19}\ \text{J}}{(500\ \text{J/s}\cdot\text{m}^2)\left[\pi(2.82\times10^{-15}\ \text{m})^2\right]} = 1.28\times10^{7}\ \text{s}$$

$$= \boxed{148\ \text{days}}$$

(b) The result for part (a) does not agree at all with the experimental observations.

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P40.23 Ultraviolet photons will be absorbed to knock electrons out of the sphere with maximum kinetic energy $K_{\max} = hf - \phi$. As the sphere loses charge, it becomes more positive relative to $V = 0$ at $r = \infty$. Eventually, the sphere will accumulate enough charge $+Q$ that the potential difference between the sphere's surface and infinity reaches the stopping potential of the photoelectrons, at which point no more electrons can escape.

$$K_{\max} = \frac{hc}{\lambda} - \phi = e\Delta V_s \quad \rightarrow \quad \Delta V_s = \frac{1}{e}\left(\frac{hc}{\lambda} - \phi\right)$$

and $$\frac{k_e Q}{r} = \Delta V_s = \frac{1}{e}\left(\frac{hc}{\lambda} - \phi\right).$$

Therefore,

$$Q = \frac{r}{ek_e}\left(\frac{hc}{\lambda} - \phi\right)$$

Solving for Q gives

$$Q = \frac{5.00\times10^{-2}\ \text{m}}{\left(1.602\times10^{-19}\ \text{C}\right)\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)} \times\left\{\left[\frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(2.998\times10^{8}\ \text{m/s}\right)}{200\times10^{-9}\ \text{m}} - 4.70\ \text{eV}\left(\frac{1.602\times10^{-19}\ \text{J}}{1\ \text{eV}}\right)\right]\right\}$$

$$= \boxed{8.34\times10^{-12}\ \text{C}}$$

P40.24 (a) The energy of photons is

$$E = \frac{hc}{\lambda} = \frac{1\ 240\ \text{nm}\cdot\text{eV}}{150\ \text{nm}} = \boxed{8.27\ \text{eV}}$$

(b) The photon energy is larger than the work function.

(c) $KE_{\max} = E - \phi = 8.27\ \text{eV} - 6.35\ \text{eV} = \boxed{1.92\ \text{eV}}$

(d) $K_{\max} = e\Delta V_s \quad \rightarrow \quad \Delta V_s = \frac{K_{\max}}{e} = \frac{1.92\ \text{eV}}{e} = \boxed{1.92\ \text{V}}$

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Section 40.3 The Compton Effect

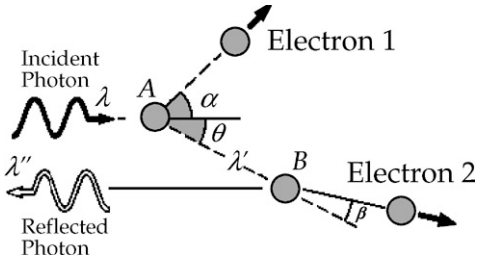
P40.25 From the Compton shift equation, the wavelength shift of the scattered x-rays is

$$\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta)$$

$$= \frac{6.63\times10^{-34}\ \text{J}\cdot\text{s}}{(9.11\times10^{-31}\ \text{kg})(3.00\times10^{8}\ \text{m/s})}(1-\cos 55.0^\circ)$$

$$= 1.03\times10^{-12}\ \text{m}\left(\frac{1\ \text{nm}}{10^{-9}\ \text{m}}\right) = \boxed{1.03\times10^{-3}\ \text{nm}}$$

P40.26 We note that $\lambda'' - \lambda = (\lambda'' - \lambda') + (\lambda' - \lambda)$.



ANS. FIG. P40.26

At A, the scattering angle is θ, and

$$\lambda' - \lambda = \frac{h}{m_e c}(1-\cos\theta)$$

At B, the scattering angle is $180^\circ - \theta$, and

$$\lambda'' - \lambda' = \frac{h}{m_e c}\left[1-\cos(180^\circ-\theta)\right] = \frac{h}{m_e c}\left[1+\cos\theta\right]$$

Therefore,

$$\lambda'' - \lambda = (\lambda'' - \lambda') + (\lambda' - \lambda)$$

$$= \frac{h}{m_e c}(1+\cos\theta) + \frac{h}{m_e c}(1-\cos\theta) = \frac{2h}{m_e c}$$

$$= \frac{2(6.63\times10^{-34}\ \text{J}\cdot\text{s})}{(9.11\times10^{-31}\ \text{kg})(3.00\times10^{8}\ \text{m/s})}$$

$$= \boxed{4.85\times10^{-12}\ \text{m}}$$

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***P40.27** This is Compton scattering through 180°:

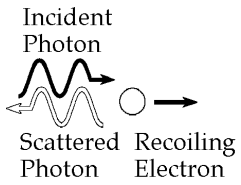
$$E_0 = \frac{hc}{\lambda_0} = \frac{(6.626\times10^{-34}\ \text{J}\cdot\text{s})(3.00\times10^{8}\ \text{m/s})}{(0.110\times10^{-9}\ \text{m})(1.60\times10^{-19}\ \text{J/eV})} = 11.3\ \text{keV}$$

$$\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta) = (2.43\times10^{-12}\ \text{m})(1-\cos 180°)$$

$$= 4.85\times10^{-12}\ \text{m}$$

$$\lambda' = \lambda_0 + \Delta\lambda = 0.115\ \text{nm},\ \text{so}\ E' = \frac{hc}{\lambda'} = 10.8\ \text{keV}.$$

By conservation of momentum for the photon-electron system,



ANS. FIG. P40.27

$$\frac{h}{\lambda_0}\hat{\mathbf{i}} = \frac{h}{\lambda'}(-\hat{\mathbf{i}}) + p_e\hat{\mathbf{i}}$$

and $$p_e = h\left(\frac{1}{\lambda_0} + \frac{1}{\lambda'}\right),$$

$$p_e = (6.626\times10^{-34}\ \text{J}\cdot\text{s})\left(\frac{(3.00\times10^{8}\ \text{m/s})/c}{1.60\times10^{-19}\ \text{J/eV}}\right)$$

$$\times\left(\frac{1}{0.110\times10^{-9}\ \text{m}} + \frac{1}{0.115\times10^{-9}\ \text{m}}\right)$$

$$= \boxed{\frac{22.1\ \text{keV}}{c}}$$

By conservation of system energy, 11.3 keV = 10.8 keV + K_e, so that $\boxed{K_e = 478\ \text{eV}}$.

Check: $$E^2 = p^2c^2 + m_e^2c^4 \ \text{or}\ \ (m_ec^2 + K_e)^2 = (pc)^2 + (m_ec^2)^2$$

$$(511\ \text{keV} + 0.478\ \text{keV})^2 = (22.1\ \text{keV})^2 + (511\ \text{keV})^2$$

$$2.62\times10^{5}\ \text{keV}^2 = 2.62\times10^{5}\ \text{keV}^2$$

P40.28 (a) and (b) From $\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta)$ we calculate the wavelength of the scattered photon. For example, at $\theta = 30°$ we have

$$\lambda' + \Delta\lambda = 120\times10^{-12}\ \text{m}$$

$$+\frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{(9.11\times10^{-31}\ \text{kg})(2.998\times10^{8}\ \text{m/s})}(1-\cos 30.0°)$$

$$= 120.3\times10^{-12}\ \text{m}$$

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The electron carries off the energy the photon loses:

$$K_e = \frac{hc}{\lambda_0} - \frac{hc}{\lambda'}$$

$$= \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(2.998\times10^{8}\ \text{m/s}\right)}{\left(1.602\times10^{-19}\ \text{J/eV}\right)} \times\left(\frac{1}{120\times10^{-12}\ \text{m}} - \frac{1}{120.3\times10^{-12}\ \text{m}}\right)$$

$$= 27.9\ \text{eV}$$

The other entries are computed similarly.

θ, degrees	0	30	60	90	120	150	180
λ', pm	120.0	120.3	121.2	122.4	123.6	124.5	124.8
K_e, eV	0	27.9	104	205	305	376	402

(c) 180°. We could answer like this: The photon imparts the greatest momentum to the originally stationary electron in a head-on collision. Here the photon recoils straight back and the electron has maximum kinetic energy.

P40.29 With $K_e = E'$ and $K_e = E_0 - E'$, we have $E' = E_0 - E' \quad \rightarrow \quad E' = \frac{E_0}{2}$.

We also have $\lambda' = \frac{hc}{E'}$; therefore, $\lambda' = \frac{hc}{E_0/2} = 2\frac{hc}{E_0} = 2\lambda_0$.

By the Compton equation,

$$\lambda' = \lambda_0 + \lambda_C(1-\cos\theta) \quad \rightarrow \quad 2\lambda_0 = \lambda_0 + \lambda_C(1-\cos\theta)$$

Therefore,

$$1-\cos\theta = \frac{\lambda_0}{\lambda_C} = \frac{0.001\,60}{0.002\,43} \quad \rightarrow \quad \theta = \boxed{70.0^\circ}$$

P40.30 (a) To compute the Compton shift, we first determine the electron's kinetic energy:

$$K = \frac{1}{2}m_e u^2 = \frac{1}{2}\left(9.11\times10^{-31}\ \text{kg}\right)\left(1.40\times10^{6}\ \text{m/s}\right)^2$$

$$= 8.93\times10^{-19}\ \text{J} = 5.58\ \text{eV}$$

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Then,

$$E_0 = \frac{hc}{\lambda_0} = \frac{1\ 240\ \text{eV}\cdot\text{nm}}{0.800\ \text{nm}} = 1\ 550\ \text{eV}$$

$$E' = E_0 - K \quad \text{and} \quad \lambda' = \frac{hc}{E'} = \frac{1\ 240\ \text{eV}\cdot\text{nm}}{1\ 550\ \text{eV} - 5.58\ \text{eV}} = 0.803\ \text{nm}$$

and the Compton shift is

$$\Delta\lambda = \lambda' - \lambda_0 = 0.002\ 89\ \text{nm} = \boxed{2.89\ \text{pm}}$$

(b) $\Delta\lambda = \lambda_C(1 - \cos\theta)$

$$\cos\theta = 1 - \frac{\Delta\lambda}{\lambda_C} = 1 - \frac{0.002\ 89\ \text{nm}}{0.002\ 43\ \text{nm}} = -0.189 \quad \rightarrow \quad \boxed{\theta = 101°}$$

P40.31 The photon has momentum $p_0 = E_0/c = h/\lambda_0$ before scattering and momentum $p' = h/\lambda'$ after scattering. The electron momentum after scattering is p_e.

(a) Conservation of momentum in the x direction gives

$$p_0 = p'\cos\theta + p_e\cos\theta$$

or $$\frac{h}{\lambda_0} = \left(\frac{h}{\lambda'} + p_e\right)\cos\theta. \qquad \textbf{[1]}$$

Conservation of momentum in the y direction gives

$$0 = p'\sin\theta - p_e\sin\theta$$

which (neglecting the trivial solution $\theta = 0$) gives

$$p_e = p' = \frac{h}{\lambda'} \qquad \textbf{[2]}$$

Substituting [2] into [1] gives

$$\frac{h}{\lambda_0} = \frac{2h}{\lambda'}\cos\theta$$

or $$\lambda' = 2\lambda_0\cos\theta. \qquad \textbf{[3]}$$

Substitute [3] into the Compton equation:

$$\lambda' - \lambda_0 = \frac{h}{m_e c}(1 - \cos\theta)$$

$$(2\lambda_0\cos\theta) - \lambda_0 = \frac{h}{m_e c}(1 - \cos\theta)$$

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Solving,

$$\left(2\lambda_0 + \frac{h}{m_e c}\right)\cos\theta = \lambda_0 + \frac{h}{m_e c}$$

$$\left(2\frac{hc}{E_0} + \frac{h}{m_e c}\right)\cos\theta = \frac{hc}{E_0} + \frac{h}{m_e c}$$

$$\frac{1}{m_e c^2 E_0}\left(2m_e c^2 + E_0\right)\cos\theta = \frac{1}{m_e c^2 E_0}\left(m_e c^2 + E_0\right)$$

$$\cos\theta = \frac{m_e c^2 + E_0}{2m_e c^2 + E_0} = \frac{0.511\ \text{MeV} + 0.880\ \text{MeV}}{2(0.511\ \text{MeV}) + 0.880\ \text{MeV}} = 0.731$$

$$\rightarrow \quad \theta = \boxed{43.0^\circ}$$

(b) Using equation [3]:

$$E' = \frac{hc}{\lambda'} = \frac{hc}{\lambda_0(2\cos\theta)} = \frac{E_0}{2\cos\theta} = \frac{0.880\ \text{MeV}}{2\cos 43.0^\circ} = \boxed{0.602\ \text{MeV}}$$

Then,

$$p' = \frac{E'}{c} = \frac{0.602\ \text{MeV}}{c} = \boxed{3.21\times10^{-22}\ \text{kg}\cdot\text{m/s}}$$

(c) From energy conservation:

$$K_e = E_0 - E' = 0.880\ \text{MeV} - 0.602\ \text{MeV} = \boxed{0.278\ \text{MeV}}$$

From equation [2],

$$p_e = p' = \frac{0.602\ \text{MeV}}{c}\left(\frac{c}{3.00\times10^{-22}\,\text{m/s}}\right)\left(\frac{1.60\times10^{-23}\,\text{J}}{1\ \text{MeV}}\right)$$

$$= \boxed{3.21\times10^{-22}\ \text{kg}\cdot\text{m/s}}$$

P40.32 The photon has momentum $p_0 = E_0/c = h/\lambda_0$ before scattering and momentum $p' = h/\lambda'$ after scattering. The electron momentum after scattering is p_e.

(a) Conservation of momentum in the x direction gives

$$p_0 = p'\cos\theta + p_e\cos\theta$$

$$\text{or} \quad \frac{h}{\lambda_0} = \left(\frac{h}{\lambda'} + p_e\right)\cos\theta. \qquad \textbf{[1]}$$

Conservation of momentum in the y direction gives

$$0 = p'\sin\theta - p_e\sin\theta$$

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which (neglecting the trivial solution $\theta = 0$) gives

$$p_e = p' = \frac{h}{\lambda'} \qquad \textbf{[2]}$$

Substituting [2] into [1] gives

$$\frac{h}{\lambda_0} = \frac{2h}{\lambda'}\cos\theta$$

or $\lambda' = 2\lambda_0 \cos\theta.$ **[3]**

Substitute [3] into the Compton equation:

$$\lambda' - \lambda_0 = \frac{h}{m_e c}(1 - \cos\theta)$$

$$(2\lambda_0 \cos\theta) - \lambda_0 = \frac{h}{m_e c}(1 - \cos\theta)$$

$$\left(2\lambda_0 + \frac{h}{m_e c}\right)\cos\theta = \lambda_0 + \frac{h}{m_e c}$$

$$\left(2\frac{hc}{E_0} + \frac{h}{m_e c}\right)\cos\theta = \frac{hc}{E_0} + \frac{h}{m_e c}$$

$$\frac{1}{m_e c^2 E_0}\left(2m_e c^2 + E_0\right)\cos\theta = \frac{1}{m_e c^2 E_0}\left(m_e c^2 + E_0\right)$$

$$\cos\theta = \frac{m_e c^2 + E_0}{2m_e c^2 + E_0} \quad \rightarrow \quad \boxed{\theta = \cos^{-1}\left(\frac{m_e c^2 + E_0}{2m_e c^2 + E_0}\right)}$$

(b) Using equation [3]:

$$E' = \frac{hc}{\lambda'} = \frac{hc}{\lambda_0(2\cos\theta)} = \frac{E_0}{2\cos\theta} = \boxed{\frac{E_0\left(2m_e c^2 + E_0\right)}{2\left(m_e c^2 + E_0\right)}}$$

Then, $$p' = \frac{E'}{c} = \boxed{\frac{E_0\left(2m_e c^2 + E_0\right)}{2c\left(m_e c^2 + E_0\right)}}.$$

(c) From energy conservation:

$$K_e = E_0 - E' = E_0 - \frac{E_0\left(2m_e c^2 + E_0\right)}{2\left(m_e c^2 + E_0\right)}$$

$$= \frac{2E_0\left(m_e c^2 + E_0\right) - E_0\left(2m_e c^2 + E_0\right)}{2\left(m_e c^2 + E_0\right)}$$

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$$K_e = \frac{\left(2E_0 m_e c^2 + 2E_0^2\right) - \left(2E_0 m_e c^2 + E_0^2\right)}{2\left(m_e c^2 + E_0\right)}$$

$$= \frac{2E_0 m_e c^2 + 2E_0^2 - 2E_0 m_e c^2 - E_0^2}{2\left(m_e c^2 + E_0\right)} = \boxed{\frac{E_0^2}{2\left(m_e c^2 + E_0\right)}}$$

From equation [2],

$$p_e = p' = \boxed{\frac{E_0\left(2m_e c^2 + E_0\right)}{2c\left(m_e c^2 + E_0\right)}}$$

P40.33 (a) From $\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta)$:

$$\Delta\lambda = \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}{\left(9.11 \times 10^{-31}\ \text{kg}\right)\left(2.998 \times 10^{8}\ \text{m/s}\right)}\left(1 - \cos 37.0^\circ\right)$$

$$= 4.89 \times 10^{-13}\ \text{m} = \boxed{4.89 \times 10^{-4}\ \text{nm}}$$

(b) $\lambda_0 = \frac{hc}{E} = \frac{1\ 240\ \text{eV} \cdot \text{nm}}{300 \times 10^{3}\ \text{eV}} = 4.13 \times 10^{-3}\ \text{nm}$

and

$$\lambda' = \lambda_0 + \Delta\lambda = 4.62 \times 10^{-12}\ \text{m} = 4.62 \times 10^{-3}\ \text{nm}$$

$$E' = \frac{hc}{\lambda'} = \frac{1\ 240\ \text{eV} \cdot \text{nm}}{4.62 \times 10^{-3}\ \text{nm}} = 2.68 \times 10^{5}\ \text{eV} = \boxed{268\ \text{keV}}$$

(c) $K_e = E_0 - E' = \boxed{31.7\ \text{keV}}$

P40.34 (a) It is, because Compton's equation and the conservation of vector momentum give three independent equations in the unknowns λ', λ_0, and u.

(b) Assuming the photon is incident along the x direction, the equations are

$$\lambda' - \lambda_0 = \frac{h}{m_e c}(1 - \cos 90.0^\circ) \quad \rightarrow \quad \lambda' = \lambda_0 + \frac{h}{m_e c} \qquad \textbf{[1]}$$

and

$$\Delta p_x = 0 \rightarrow \frac{h}{\lambda_0} = \gamma m_e u \cos 20.0^\circ$$

$$\Delta p_y = 0 \rightarrow \frac{h}{\lambda'} = \gamma m_e u \sin 20.0^\circ$$

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Dividing the latter two equations gives

$$\frac{\lambda_0}{\lambda'} = \tan 20.0° \qquad \textbf{[2]}$$

Substituting equation [2] into equation [1] gives

$$\lambda' = \lambda' \tan 20.0° + \frac{h}{m_e c}$$

$$\lambda' = \frac{h}{m_e c(1 - \tan 20.0°)} = \frac{hc}{m_e c^2 (1 - \tan 20.0°)}$$

$$= \frac{1240 \text{ eV} \cdot \text{nm}}{(0.511 \times 10^6 \text{ eV})(1 - \tan 20.0°)}$$

$$= 3.82 \times 10^{-3} \text{ nm} = 3.82 \times 10^{-12} \text{ m} = \boxed{3.82 \text{ pm}}$$

P40.35 We treat the electron non-relativistically because

$$\frac{u}{c} = \frac{2.18 \times 10^6 \text{ m/s}}{3.00 \times 10^3 \text{ m/s}} = 0.007\ 27 < 0.01$$

The electron's final kinetic energy is

$$K_f = \frac{1}{2} m_e u^2 .$$

This is the energy lost by the photon:

$$\Delta E = hf_0 - hf' = \frac{hc}{\lambda_0} - \frac{hc}{\lambda'} = K_f \qquad \textbf{[1]}$$

From the Compton equation, we have

$$\Delta\lambda = \lambda' - \lambda_0 = \frac{h}{m_e c}(1 - \cos\theta) \qquad \textbf{[2]}$$

$$\lambda' = \lambda_0 + \frac{h}{m_e c}(1 - \cos\theta) \qquad \textbf{[3]}$$

Substitute [2] and [3] into [1]:

$$K_f = \frac{hc}{\lambda_0} - \frac{hc}{\lambda'} = \frac{(\lambda_0 - \lambda')hc}{\lambda_0 \lambda'} = \frac{h}{m_e c}(1 - \cos\theta)\frac{hc}{\lambda_0 \lambda'}$$

$$K_f = \frac{h^2 c(1 - \cos\theta)}{m_e c \lambda_0 \left[\lambda_0 + \frac{h}{m_e c}(1 - \cos\theta)\right]}$$

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Solving,

$$m_e c\lambda_0\left[\lambda_0 + \frac{h}{m_e c}(1-\cos\theta)\right] = \frac{h^2 c}{K_f}(1-\cos\theta)$$

$$m_e c\lambda_0^2 + h(1-\cos\theta)\lambda_0 - \frac{h^2 c}{K_f}(1-\cos\theta) = 0$$

(a) Solve for λ_0:

$$\lambda_0 = \frac{h(1-\cos\theta) \pm \sqrt{\left[h(1-\cos\theta)\right]^2 - 4\left(m_e c\right)\left[-\frac{h^2 c}{K_f}(1-\cos\theta)\right]}}{2m_e c}$$

$$\lambda_0 = \frac{h(1-\cos\theta) \pm \sqrt{\left[h(1-\cos\theta)\right]^2 + \left[\frac{4h^2 m_e c^2}{\frac{1}{2}m_e u^2}(1-\cos\theta)\right]}}{2m_e c}$$

$$\lambda_0 = \frac{h(1-\cos\theta) \pm \sqrt{\left[h(1-\cos\theta)\right]^2 + \left[\frac{8h^2 c^2}{u^2}(1-\cos\theta)\right]}}{2m_e c}$$

$$= \frac{h(1-\cos\theta)}{2m_e c}\left\{1 \pm \sqrt{1+\left[\frac{8c^2}{u^2(1-\cos\theta)}\right]}\right\}$$

Only the positive answer is physical:

$$\lambda_0 = \frac{h(1-\cos\theta)}{2m_e c}\left\{1 + \sqrt{1+\left[\frac{8c^2}{u^2(1-\cos\theta)}\right]}\right\}$$

$$= \frac{\left(6.63\times10^{-34}\ \text{J}\cdot\text{s}\right)(1-\cos 17.4^\circ)}{2\left(9.11\times10^{-31}\ \text{kg}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}$$

$$\times\left\{1 + \sqrt{1+\left[\frac{8\left(3.00\times10^{8}\ \text{m/s}\right)^2}{\left(2.18\times10^{6}\ \text{m/s}\right)^2(1-\cos 17.4^\circ)}\right]}\right\}$$

$$= 1.01\times10^{-10}\ \text{m} = \boxed{0.101\ \text{nm}}$$

(b) From [3],

$$\lambda' = \lambda_0 + \frac{h}{m_e c}(1-\cos\theta)$$

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Substituting,

$$\lambda' = \frac{h(1-\cos\theta)}{2m_e c}\left\{1+\sqrt{1+\left[\frac{8c^2}{u^2(1-\cos\theta)}\right]}\right\}+\frac{h}{m_e c}(1-\cos\theta)$$

$$= \frac{h}{m_e c}(1-\cos\theta)\left\{\frac{3}{2}+\frac{1}{2}\sqrt{1+\left[\frac{8c^2}{u^2(1-\cos\theta)}\right]}\right\}$$

$$= 1.0116\times10^{-10}\ \text{m}$$

The electron scattering angle is ϕ. By conservation of momentum in the transverse direction:

$$0 = \frac{h}{\lambda'}\sin\theta - m_e u\sin\phi \quad\rightarrow\quad \sin\phi = \frac{h}{\lambda' m_e u}\sin\theta$$

$$\phi = \sin^{-1}\left(\frac{h}{\lambda' m_e u}\sin\theta\right)$$

$$= \sin^{-1}\left(\frac{6.63\times10^{-34}\ \text{J}\cdot\text{s}}{\lambda'\left(9.11\times10^{-31}\ \text{kg}\right)\left(2.18\times10^{6}\ \text{m/s}\right)}\sin 17.4^\circ\right)$$

$$= \boxed{80.7^\circ}$$

P40.36 Maximum energy loss appears as maximum increase in wavelength, which occurs for scattering angle 180°. Then,

$$\Delta\lambda = \left(\frac{h}{mc}\right)(1-\cos 180^\circ) = \frac{2h}{mc}$$

where m is the mass of the target particle. The fractional energy loss is

$$\frac{E_0 - E'}{E_0} = \frac{hc/\lambda_0 - hc/\lambda'}{hc/\lambda_0} = \frac{\lambda' - \lambda_0}{\lambda'} = \frac{\Delta\lambda}{\lambda_0 + \Delta\lambda} = \frac{2h/mc}{\lambda_0 + 2h/mc}$$

Further, $\lambda_0 = \dfrac{hc}{E_0}$, so $\dfrac{E_0 - E'}{E_0} = \dfrac{2h/mc}{hc/E_0 + 2h/mc} = \dfrac{2E_0}{mc^2 + 2E_0}$.

(a) For scattering from a free electron, $mc^2 = 0.511$ MeV, so

$$\frac{E_0 - E'}{E_0} = \frac{2(0.511\ \text{MeV})}{0.511\ \text{MeV} + 2(0.511\ \text{MeV})} = \boxed{0.667}$$

(b) For scattering from a free proton, $mc^2 = 938$ MeV, and

$$\frac{E_0 - E'}{E_0} = \frac{2(0.511\ \text{MeV})}{938\ \text{MeV} + 2(0.511\ \text{MeV})} = \boxed{0.001\,09}$$

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Section 40.4 The Nature of Electromagnetic Waves

P40.37 With photon energy $E = hf = 10.0$ eV, a photon would have

$$f = \frac{E}{h} = \frac{10.0\left(1.602\times10^{-19}\ \text{J}\right)}{6.626\times10^{-34}\ \text{J}\cdot\text{s}} = 2.42\times10^{15}\ \text{Hz}$$

and

$$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\ \text{m/s}}{2.41\times10^{15}\ \text{Hz}} = 124\ \text{nm}$$

To have photon energy 10 eV or greater, according to this definition, ionizing radiation is the ultraviolet light, x-rays, and γ rays with wavelength shorter than 124 nm; that is, with frequency higher than 2.42×10^{15} Hz.

P40.38 The photon energy is

$$E = \frac{hc}{\lambda} = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}{633\times10^{-9}\ \text{m}} = 3.14\times10^{-19}\ \text{J}$$

The power carried by the beam is

$$\left(2.00\times10^{18}\ \text{photons/s}\right)\left(3.14\times10^{-19}\ \text{J/photon}\right) = 0.628\ \text{W}$$

Its intensity is the average Poynting vector

$$I = S_{\text{avg}} = \frac{P}{\pi r^2} = \frac{0.628\ \text{W}}{\pi\left(\dfrac{1.75\times10^{-3}\ \text{m}}{2}\right)^2} = 2.61\times10^{5}\ \text{W/m}^2$$

(a) To find the electric field, we use

$$S_{\text{avg}} = \frac{1}{\mu_0}E_{\text{rms}}B_{\text{rms}} = \frac{E_{\text{max}}^2}{2\mu_0 c}$$

Solving,

$$E_{\text{max}} = \left(2\mu_0 c S_{\text{avg}}\right)^{1/2}$$

$$= \left[2\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times10^{8}\ \text{m/s}\right)\times\left(2.61\times10^{5}\ \text{W/m}^2\right)\right]^{1/2}$$

$$= 1.40\times10^{4}\ \text{N/C} = \boxed{14.0\ \text{kV/m}}$$

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(b) $B_{\max} = \dfrac{E_{\max}}{c} = \dfrac{1.40 \times 10^4 \text{ N/C}}{3.00 \times 10^8 \text{ m/s}} = 4.68 \times 10^{-5} \text{ T} = \boxed{46.8\ \mu\text{T}}$

(c) Each photon carries momentum $\dfrac{E}{c}$. The beam transports momentum at the rate $\dfrac{P}{c}$. It imparts momentum to a perfectly reflecting surface at the rate

$$\frac{2P}{c} = \text{force} = \frac{2(0.628 \text{ W})}{3.00 \times 10^8 \text{ m/s}} = 4.19 \times 10^{-9} \text{ N} = \boxed{4.19 \text{ nN}}$$

(d) The block of ice absorbs energy $mL = P\Delta t$ melting

$$m = \frac{P\Delta t}{L} = \frac{(0.628 \text{ W})\left[1.50(3\,600 \text{ s})\right]}{3.33 \times 10^5 \text{ J/kg}} = 1.02 \times 10^{-2} \text{ kg} = \boxed{10.2 \text{ g}}$$

Section 40.5 The Wave Properties of Particles

P40.39 (a) $p = \dfrac{h}{\lambda} = \dfrac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{4.00 \times 10^{-7} \text{ m}} = \boxed{1.66 \times 10^{-27} \text{ kg} \cdot \text{m/s}}$

(b) From $p = m_e u$,

$$u = \frac{p}{m_e} = \frac{1.66 \times 10^{-27} \text{ kg} \cdot \text{m/s}}{9.11 \times 10^{-31} \text{ kg}} = 1.82 \times 10^3 \text{ m/s} = \boxed{1.82 \text{ km/s}}$$

P40.40 (a) Electron: $\lambda = \dfrac{h}{p}$ and $K = \dfrac{1}{2} m_e u^2 = \dfrac{m_e^2 u^2}{2m_e} = \dfrac{p^2}{2m_e}$,

so $p = \sqrt{2m_e K}$

and $\lambda = \dfrac{h}{\sqrt{2m_e K}} = \dfrac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{\sqrt{2(9.11 \times 10^{-31} \text{ kg})(3.00)(1.60 \times 10^{-19} \text{ J})}}$

$= 7.09 \times 10^{-10} \text{ m} = \boxed{0.709 \text{ nm}}$

(b) Photon: $\lambda = \dfrac{c}{f}$ and $E = hf$, so $f = \dfrac{E}{h}$,

and $\lambda = \dfrac{hc}{E} = \dfrac{1240 \text{ eV} \cdot \text{nm}}{3.00 \text{ eV}} = \boxed{413 \text{ nm}}$.

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P40.41 Since the de Broglie wavelength is $\lambda = \frac{h}{p}$, the electron momentum is:

$$p = \frac{h}{\lambda} \approx \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}{1.00 \times 10^{-11}\ \text{m}} = 6.626 \times 10^{-23}\ \text{kg} \cdot \text{m/s}$$

(a) For electrons, the relativistic answer is more precisely correct. Suppressing units,

$$K_e = \sqrt{p^2c^2 + \left(m_e c^2\right)^2} - m_e c^2 = \sqrt{\left(pc\right)^2 + \left(m_e c^2\right)^2} - m_e c^2$$

$$= \sqrt{\left[\left(6.626 \times 10^{-23}\right)\left(2.998 \times 10^{8}\right)\left(\frac{1\ \text{MeV}}{1.602 \times 10^{-13}\ \text{J}}\right)\right]^2 + (0.511)^2}$$

$$-0.511$$

$$= 0.014\ 8\ \text{MeV} = \boxed{14.8\ \text{keV}}$$

or, ignoring relativistic correction,

$$K_e = \frac{p^2}{2m_e} = \frac{\left(6.626 \times 10^{-23}\right)^2}{2\left(9.11 \times 10^{-31}\right)} \left(\frac{1\ \text{keV}}{1.602 \times 10^{-16}\ \text{J}}\right) = \boxed{15.1\ \text{keV}}$$

(b) For photons (suppressing units):

$$E_\gamma = pc = \left(6.626 \times 10^{-23}\right)\left(2.998 \times 10^{8}\right)\left(\frac{1\ \text{keV}}{1.602 \times 10^{-16}\ \text{J}}\right)$$

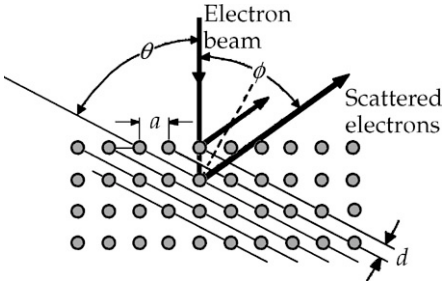
$$= \boxed{124\ \text{keV}}$$

P40.42 The de Broglie wavelength of the proton is

$$\lambda = \frac{h}{p} = \frac{h}{mu} = \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}{\left(1.67 \times 10^{-27}\ \text{kg}\right)\left(1.00 \times 10^{6}\ \text{m/s}\right)}$$

$$= \boxed{3.97 \times 10^{-13}\ \text{m}}$$

P40.43 Refer to Figure P40.43 (or ANS. FIG. P40.43). For Bragg reflection, the angle θ is measured from the reflecting plane to the incident beam, as shown in Figure 38.23. Angle ϕ is measured from the incident beam to the reflected (scattered) beam. The law of reflection applies relative to the normal to the plane (the dashed line), so the angles of incidence and reflection are equal to $\phi/2$. The angle between the reflecting plane and



ANS. FIG. P40.43(a)

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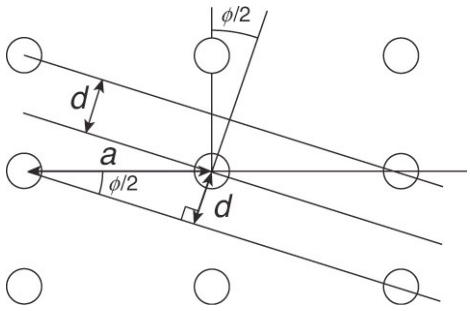
the normal is 90°, so

$$\theta + \frac{\phi}{2} = 90°$$

From the condition for Bragg reflection, we have

$$m\lambda = 2d\sin\theta = 2d\sin\left(90° - \frac{\phi}{2}\right)$$

$$= 2d\cos\left(\frac{\phi}{2}\right)$$



ANS. FIG. P40.43(b)

The vertical beam is incident along the normal to the horizontal lattice planes which contain atoms that are separated by distance a, and the reflecting lattice planes form the angle $\phi/2$ with the horizontal planes because the normal to the reflecting planes forms the angle $\phi/2$ with the vertical beam. Therefore, the spacing of the reflecting lattice planes is $d = a\sin\left(\frac{\phi}{2}\right)$.

Thus, for the first maximum, with $m = 1$,

$$\lambda = 2\left[a\sin\left(\frac{\phi}{2}\right)\right]\cos\left(\frac{\phi}{2}\right) = a\sin\phi$$

We know that

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e K}}$$

$$= \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{\sqrt{2\left(9.11\times10^{-31}\ \text{kg}\right)\left(54.0\times1.60\times10^{-19}\ \text{J}\right)}} = 1.67\times10^{-10}\ \text{m}$$

Therefore, the lattice spacing is

$$a = \frac{\lambda}{\sin\phi} = \frac{1.67\times10^{-10}\ \text{m}}{\sin 50.0°} = 2.18\times10^{-10} = \boxed{0.218\ \text{nm}}$$

P40.44 (a) $\lambda \sim 10^{-14}$ m or less, so $p = \frac{h}{\lambda} \sim \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{10^{-14}\ \text{m}} \approx 10^{-19}\ \text{kg}\cdot\text{m/s}$ or more. The energy of the electron is, suppressing units,

$$E = \sqrt{p^2c^2 + m_e^2c^4}$$

$$\sim \sqrt{\left(10^{-19}\right)^2\left(3\times10^8\right)^2 + \left(9\times10^{-31}\right)^2\left(3\times10^8\right)^4}$$

or $E \sim 10^{-11}\ \text{J} \sim 10^8\ \text{eV}$ or more

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so that

$$K = E - m_e c^2 \sim 10^8 \text{ eV} - \left(0.5 \times 10^6 \text{ eV}\right) \boxed{\sim 10^8 \text{ eV}} \text{ or more}$$

(b) If the nucleus contains ten protons, the electric potential energy of the electron-nucleus system would be

$$U_e = \frac{k_e q_1 q_2}{r} \sim \frac{\left(9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right)\left[10\left(1.60 \times 10^{-19} \text{ C}\right)\right](-e)}{0.5 \times 10^{-14} \text{ m}}$$

$$\boxed{\sim -10^6 \text{ eV}}$$

(c) With its $K + U_e \sim 10^8 \text{ eV} >> 0$, the electron could not be confined to the nucleus.

P40.45 (a) From $E = \gamma mc^2$,

$$\gamma = \frac{E}{mc^2} = \frac{20\ 000 \text{ MeV}}{0.511 \text{ MeV}} = \boxed{3.91 \times 10^4}$$

(b) We find the momentum of the particle from

$$pc = \left[E^2 - \left(mc^2\right)^2\right]^{1/2} = \left[(20\ 000 \text{ MeV})^2 - (0.511 \text{ MeV})^2\right]$$
$$= 20.0 \text{ GeV}$$

Then,

$$p = \boxed{20.0 \text{ GeV}/c = 1.07 \times 10^{-17} \text{ kg} \cdot \text{m/s}}$$

(c) The electron's wavelength is

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{1.07 \times 10^{-17} \text{ kg} \cdot \text{m/s}} = \boxed{6.21 \times 10^{-17} \text{ m}}$$

(d) The wavelength is two orders of magnitude smaller than the size of the nucleus.

P40.46 Given the assumption in the problem statement, for significant diffraction to occur, we must have

$$w \leq 10\lambda = 10\left(\frac{h}{p}\right) = 10\left(\frac{h}{mu}\right)$$

where u is the speed of the student as he passes through the doorway. The variable we do not know here is the speed u, so let's solve for it:

$$u \leq 10\left(\frac{h}{mw}\right)$$

This expression will give the upper limit to the speed of the student.

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Substitute numerical values:

$$u \le 10\left[\frac{6.626 \times 10^{-34}\ \text{J}\cdot\text{s}}{(80\ \text{kg})(0.75\ \text{m})}\right] = 1.1 \times 10^{-34}\ \text{m/s}$$

This is an extremely low velocity. It is impossible for the student to walk this slowly. At this speed, if the thickness of the wall in which the door is built is 15 cm, the time interval required for the student to pass through the door is 1.4×10^{33} s, which is 10^{15} times the age of the Universe.

P40.47 (a) For the electron,

$$K = (\gamma - 1)m_e c^2 \qquad \text{and} \qquad \lambda = \frac{h}{p} = \frac{h}{\gamma m_e u}$$

For the photon,

$$E_{\text{ph}} = K \qquad \text{and} \qquad \lambda_{\text{ph}} = \frac{c}{f} = \frac{ch}{E} = \frac{ch}{K} = \frac{ch}{(\gamma - 1)m_e c^2}$$

Then the ratio is

$$\frac{\lambda_{\text{ph}}}{\lambda} = \frac{ch}{(\gamma - 1)m_e c^2}\frac{\gamma m_e u}{h} = \boxed{\frac{\gamma}{\gamma - 1}\frac{u}{c},\ \text{where}\ \gamma = \frac{1}{\sqrt{1 - u^2/c^2}}}$$

(b) For $u = 0.900c$,

$$\frac{\lambda_{\text{ph}}}{\lambda} = \frac{1}{\sqrt{1-(0.900)^2}}\frac{(0.900)}{\left[\left(1/\sqrt{1-(0.900)^2}\right)-1\right]} = \boxed{1.60}$$

(c) The ratio for a particular particle speed does not depend on the particle mass: There would be boxed{no change.}

(d) For $u = 0.001\ 00c$,

$$\frac{\lambda_{\text{ph}}}{\lambda} = \frac{1}{\sqrt{1-(0.001\ 00)^2}}\frac{(0.001\ 00)}{\left[\left(1/\sqrt{1-(0.001\ 00)^2}\right)-1\right]} = \boxed{2.00 \times 10^3}$$

(e) As $\frac{u}{c} \to 1$, $\gamma \to \infty$ and $\gamma - 1$ becomes nearly equal to γ. Then,

$$\frac{\lambda_\gamma}{\lambda_m} \to \frac{\gamma}{\gamma}(1) = \boxed{1}$$

(f) As $\frac{u}{c} \to 0$, $\left(1 - \frac{u^2}{c^2}\right)^{-1/2} - 1 \approx 1 - \left(-\frac{1}{2}\right)\frac{u^2}{c^2} - 1 = \frac{1}{2}\frac{u^2}{c^2}$ and

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$$\frac{\lambda_\gamma}{\lambda_m} \to 1\frac{u/c}{(1/2)\left(u^2/c^2\right)} = \frac{2c}{u} \to \boxed{\infty}$$

P40.48 (a) $E^2 = p^2c^2 + m^2c^4$ with $E = hf$,

$$p = \frac{h}{\lambda} \quad \text{and} \quad mc = \frac{h}{\lambda_C}$$

Substituting, we find that

$$h^2 f^2 = \frac{h^2c^2}{\lambda^2} + \frac{h^2c^2}{\lambda_C^2} \quad \text{and} \quad \left(\frac{f}{c}\right)^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda_C^2}$$

(b) No. For a photon, $\frac{f}{c} = \frac{1}{\lambda}$. The third term $\frac{1}{\lambda_C}$ in the equation above for particles with mass shows that they will always have a different frequency from photons of the same wavelength.

Section 40.6 A New Model: The Quantum Particle

P40.49 (a) The particle is freely moving, so we attribute no potential energy to it. Its energy is

$$E = K = \frac{1}{2}mu^2 = hf = \left(\frac{h}{2\pi}\right)(2\pi f) = \hbar\omega$$

For its momentum we have

$$p = mu = \frac{h}{\lambda} = \left(\frac{h}{2\pi}\right)\left(\frac{2\pi}{\lambda}\right) = \hbar k$$

Thus,

$$\omega = \frac{K}{\hbar} \quad \text{and} \quad k = \frac{p}{\hbar}$$

Then the phase speed is

$$v_{\text{phase}} = f\lambda = \left(\frac{mu^2}{2h}\right)\left(\frac{h}{mu}\right) = \boxed{\frac{u}{2}}$$

(b) We see that the phase speed is only one-half of the experimentally measurable speed u at which the quantum particle transports mass, energy, and momentum. In the textbook's Active Figure 28.17, individual wave crests would move forward more slowly than their envelope moves forward, so individual crests would appear to move backward relative to the packet containing them.

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P40.50 As a bonus, we begin by proving that the phase speed $v_p = \dfrac{\omega}{k}$ is not the speed of the particle.

$$v_p = \frac{\omega}{k} = \frac{\sqrt{p^2c^2 + m^2c^4}\,\hbar}{\hbar\gamma mu} = \frac{\sqrt{\gamma^2 m^2 u^2 c^2 + m^2 c^4}}{\sqrt{\gamma^2 m^2 u^2}}$$

$$= c\sqrt{1 + \frac{c^2}{\gamma^2 u^2}} = c\sqrt{1 + \frac{c^2}{u^2}\left(1 - \frac{u^2}{c^2}\right)} = c\sqrt{1 + \frac{c^2}{u^2} - 1} = \frac{c^2}{u}$$

In fact, the phase speed is larger than the speed of light! A point of constant phase in the wave function carries no mass, no energy, and no information.

Now for the group speed:

$$v_g = \frac{d\omega}{dk} = \frac{d(\hbar\omega)}{d(\hbar k)} = \frac{dE}{dp} = \frac{d}{dp}\sqrt{m^2c^4 + p^2c^2}$$

$$= \frac{1}{2}\left(m^2c^4 + p^2c^2\right)^{-1/2}\left(0 + 2pc^2\right) = \sqrt{\frac{p^2c^4}{p^2c^2 + m^2c^4}}$$

$$= c\sqrt{\frac{\gamma^2 m^2 u^2}{\gamma^2 m^2 u^2 + m^2 c^2}}$$

$$= c\sqrt{\frac{u^2/\left(1 - u^2/c^2\right)}{u^2/\left(1 - u^2/c^2\right) + c^2}} = c\sqrt{\frac{u^2/\left(1 - u^2/c^2\right)}{\left(u^2 + c^2 - u^2\right)/\left(1 - u^2/c^2\right)}} = u$$

It is this speed at which mass, energy, and momentum are transported.

Section 40.7 The Double-Slit Experiment Revisited

P40.51 (a) $\lambda = \dfrac{h}{mu} = \dfrac{6.626 \times 10^{-34}\ \text{J}\cdot\text{s}}{\left(1.67 \times 10^{-27}\ \text{kg}\right)\left(0.400\ \text{m/s}\right)} = 9.92 \times 10^{-7}\ \text{m} = \boxed{992\ \text{nm}}$

(b) For destructive interference in a multiple-slit experiment,

$d\sin\theta = \left(m + \dfrac{1}{2}\right)\lambda$, with $m = 0$ for the first minimum. Also,

$\dfrac{y}{L} = \tan\theta \approx \sin\theta = \left(\dfrac{1}{2}\right)\dfrac{\lambda}{d}$, so

$$y = L\tan\theta = \frac{\lambda L}{2d} = \frac{\left(9.92 \times 10^{-7}\ \text{m}\right)(10.0\ \text{m})}{2\left(1.00 \times 10^{-3}\ \text{m}\right)} = \boxed{4.96\ \text{mm}}$$

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(c) No; there is no way to identify the slit through which the neutron passed. Even if one neutron at a time is incident on the pair of slits, an interference pattern still develops on the detector array. Therefore, each neutron in effect passes through both slits.

P40.52 We find the speed of each electron from energy conservation in the firing process:

$$E = 0 = K_f + U_f = \frac{1}{2}mu^2 - \text{eV}$$

$$u = \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2\left(1.60\times10^{-19}\text{ C}\right)(45.0\text{ V})}{9.11\times10^{-31}\text{ kg}}} = 3.98\times10^{6}\text{ m/s}$$

The time of flight is

$$\Delta t = \frac{\Delta x}{u} = \frac{0.280\text{ m}}{3.98\times10^{6}\text{ m/s}} = 7.04\times10^{-8}\text{ s}$$

The current when electrons are 28 cm apart is

$$I = \frac{q}{t} = \frac{e}{\Delta t} = \frac{1.60\times10^{-19}\text{ C}}{7.04\times10^{-8}\text{ s}} = \boxed{2.27\times10^{-12}\text{ A}}$$

P40.53 Consider the first bright band away from the center:

$$d\sin\theta = m\lambda$$

$$\left(0.060\,0\times10^{-6}\text{ m}\right)\sin\left[\tan^{-1}\left(\frac{0.400\times10^{-3}\text{ m}}{20.0\times10^{-2}\text{ m}}\right)\right]$$

$$= (1)\lambda = 1.20\times10^{-10}\text{ m}$$

And since $\lambda = \frac{h}{p} = \frac{h}{m_e u}$, so $m_e u = \frac{h}{\lambda}$,

and

$$K = \frac{1}{2}m_e u^2 = \frac{m_e^2 u^2}{2m_e} = \frac{h^2}{2m_e\lambda^2} = e\Delta V \qquad \rightarrow \qquad \Delta V = \frac{h^2}{2em_e\lambda^2}$$

Therefore,

$$\Delta V = \frac{\left(6.626\times10^{-34}\text{ J}\cdot\text{s}\right)^2}{2\left(1.60\times10^{-19}\text{ C}\right)\left(9.11\times10^{-31}\text{ kg}\right)\left(1.20\times10^{-10}\text{ m}\right)^2}$$

$$= \boxed{105\text{ V}}$$

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Section 40.8 The Uncertainty Principle

P40.54 (a) The uncertainty principle states $\Delta p \Delta x = m \Delta u \Delta x \geq \frac{\hbar}{2}$, so

$$\Delta u \geq \frac{h}{4\pi m \Delta x} = \frac{2\pi \text{ J} \cdot \text{s}}{4\pi (2.00 \text{ kg})(1.00 \text{ m})} = \boxed{0.250 \text{ m/s}}$$

(b) The duck might move by (0.250 m/s)(5.00 s) = 1.25 m. With an original position uncertainty of 1.00 m, we can think of Δx growing to $1.00 \text{ m} + 1.25 \text{ m} = \boxed{2.25 \text{ m}}$.

P40.55 The uncertainty principle states $\Delta x \Delta p_x \geq \frac{\hbar}{2}$, where $\Delta p_x = m \Delta u$ and $\hbar = h/2\pi$.

Both the electron and bullet have a velocity uncertainty

$$\Delta u = (0.000\ 100)(500 \text{ m/s}) = 0.050\ 0 \text{ m/s}$$

For the electron, the minimum uncertainty in position is

$$\Delta x = \frac{h}{4\pi m \Delta u} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{4\pi (9.11 \times 10^{-31} \text{ kg})(0.050\ 0 \text{ m/s})} = \boxed{1.16 \text{ mm}}$$

For the bullet,

$$\Delta x = \frac{h}{4\pi m \Delta u} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{4\pi (0.020\ 0 \text{ kg})(0.050\ 0 \text{ m/s})} = \boxed{5.28 \times 10^{-32} \text{ m}}$$

P40.56 The momentum of the block is $p = mv$, and if the mass is known precisely, the uncertainty in the momentum is $\Delta p = m \Delta v$. From the uncertainty principle, $\Delta x \, \Delta p_x \geq \hbar/2$, so if there is an uncertainty of $\Delta x = 0.150 \text{ cm} = 1.50 \times 10^{-3} \text{ m}$ in the position of the particle, the minimum uncertainty in its speed is

$$(\Delta v_x)_{\min} = \frac{(\Delta p_x)_{\min}}{m} = \frac{h}{4\pi m (\Delta x)_{\max}}$$

$$= \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{4\pi (0.500 \text{ kg})(1.50 \times 10^{-3} \text{ m})} = \boxed{7.03 \times 10^{-32} \text{ m/s}}$$

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P40.57 The maximum time one can use in measuring the energy of the particle is equal to the lifetime of the particle, or $\Delta t_{\max} \approx 2\ \mu\text{s}$. One form of the uncertainty principle is $\Delta E\,\Delta t \geq \hbar/2$. Thus, the minimum uncertainty one can have in the measurement of a muon's energy is

$$\Delta E_{\min} = \frac{h}{4\pi\,\Delta t_{\max}} = \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{4\pi\left(2\times10^{-6}\ \text{s}\right)} = \boxed{3\times10^{-29}\ \text{J} \approx 2\times10^{-10}\ \text{eV}}$$

P40.58 Assume the rifle is firing horizontally and let the distance between the rifle and the target be *L*. The uncertainty in the vertical position of the particle as it leaves the end of the rifle is $\Delta y = 2.00$ mm. The uncertainty principle will allow us to approximate the uncertainty in the vertical momentum of the particles (ignoring gravitational acceleration):

$$\Delta y \Delta p_y \geq \frac{\hbar}{2} \qquad \rightarrow \qquad \Delta p_y \geq \frac{\hbar}{2\Delta y}$$

The time interval for the particle to reach the screen is, from the particle under constant velocity model,

$$\Delta t = \frac{L}{v_x}$$

During this time interval, again from the particle under constant velocity model, the particle moves in the vertical direction by a distance (again ignoring gravitational effects)

$$\Delta y_t = v_y \Delta t = v_y \frac{L}{v_x} = p_y \frac{L}{p_x}$$

where Δy_t is the vertical distance though which the particle moves when it arrives at the target and p_y is the vertical momentum of the particle. Because the particles begin with zero vertical momentum, let's assume that the vertical momentum of the particles is on the order of the uncertainty in the vertical momentum. Then,

$$\Delta y_t \approx \frac{\hbar}{2\Delta y}\ \frac{L}{p_x}$$

What we don't know in this expression is the distance *L*, so let's solve for it:

$$L \approx \frac{2p_x \Delta y \Delta y_t}{\hbar}$$

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Substitute numerical values:

$$L \approx \frac{2(0.001\ 00\ \text{kg})(100\ \text{m/s})(0.002\ 00\ \text{m})(0.010\ 0\ \text{m})}{1.055 \times 10^{-34}\ \text{J}\cdot\text{s}}$$

$$\approx 4 \times 10^{28}\ \text{m}$$

According to Table 1.1, this distance is two orders of magnitude larger than the distance from the Earth to the most remote known quasar. In conclusion, then, for rifles fired at targets at reasonable distances away, a spread of 1.00 cm *due to the uncertainty principle* would be impossible.

P40.59 With $\Delta x = 1 \times 10^{-14}\ \text{m},$ the uncertainty principle requires

$$\Delta p_x \geq \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34}\ \text{J}\cdot\text{s}}{2\left(1 \times 10^{-14}\ \text{m}\right)} = 5.3 \times 10^{-21}\ \text{kg}\cdot\text{m/s}$$

The average momentum of the particle bound in a stationary nucleus is zero. The uncertainty in momentum measures the standard deviation of the momentum, so we take $p \approx 5.3 \times 10^{-21}\ \text{kg}\cdot\text{m/s}$.

For an electron, the non-relativistic approximation $p = m_e u$ would predict $u \approx 6 \times 10^{9}\ \text{m/s}$, which is impossible because u cannot be greater than c. Thus, a better solution would be to use

$$E = \left[\left(m_e c^2\right)^2 + \left(pc\right)^2\right]^{1/2} \approx 9.9\ \text{MeV} = \gamma m_e c^2$$

to find the speed (with $m_e c^2 = 0.511$ MeV):

$$\gamma \approx 19.4 = \frac{1}{\sqrt{1 - u^2/c^2}} \quad \text{so} \quad u \approx 0.998\ 67c$$

For a proton,

$$u = \frac{p}{m} = \frac{5.3 \times 10^{-21}\ \text{kg}\cdot\text{m/s}}{1.67 \times 10^{-27}\ \text{kg}} = 3.2 \times 10^{6}\ \text{m/s} = 0.011c$$

about one-hundredth the speed of light.

Additional Problems

P40.60 From each wavelength we find the corresponding frequency using the relation $\lambda f = c$, where c is the speed of light:

For $\lambda_1 = 588 \times 10^{-9}\ \text{m}$ $f_1 = \frac{c}{\lambda_1} = 5.10 \times 10^{14}\ \text{Hz}$

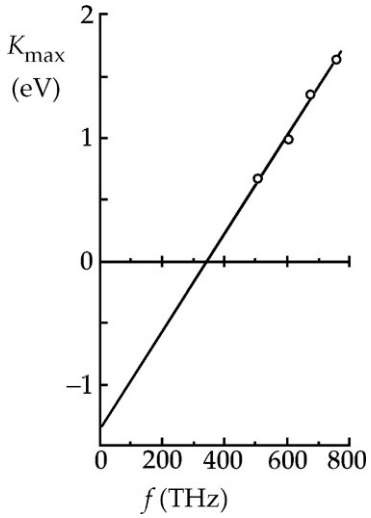
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For $\lambda_2 = 505 \times 10^{-9}\ \text{m}$ $f_2 = 5.94 \times 10^{14}\ \text{Hz}$

$\lambda_3 = 445 \times 10^{-9}\ \text{m}$ $f_3 = 6.74 \times 10^{14}\ \text{Hz}$

$\lambda_4 = 399 \times 10^{-9}\ \text{m}$ $f_4 = 7.52 \times 10^{14}\ \text{Hz}$

(a) We plot each point on an energy versus frequency graph, as shown in ANS. FIG. P40.60. We extend a straight line through the set of 4 points, as far as the negative y intercept.



ANS. FIG. P40.60

(b) Our basic equation is $K_{max} = hf - \phi$. Therefore, an experimental value for Planck's constant is the slope of the K-f graph, which can be found from a least-squares fit or from reading the graph as:

$$h_{exp} = \frac{\text{Rise}}{\text{Run}} = \frac{1.25\ \text{eV} - 0.25\ \text{eV}}{6.5 \times 10^{14}\ \text{Hz} - 4.0 \times 10^{14}\ \text{Hz}}$$

$$= 4.0 \times 10^{-15}\ \text{eV} \cdot \text{s} = \boxed{6.4 \times 10^{-34}\ \text{J} \cdot \text{s}}$$

From the scatter of the data points on the graph, we estimate the uncertainty of the slope to be about 3%. Thus we choose to show two significant figures in writing the experimental value of Planck's constant.

(c) Again from the linear equation $K_{max} = hf - \phi$, the work function for the metal surface is the negative of the y-intercept of the graph, so

$$\phi_{exp} = -(-1.4\ \text{eV}) = \boxed{1.4\ \text{eV}}$$

Based on the range of slopes that appear to fit the data, the estimated uncertainty of the work function is 5%.

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***P40.61** From the circular path the electrons follow in the magnetic field, the magnetic force is centripetal,

$$F = ma: \qquad evB = \frac{m_e v^2}{R} \quad \rightarrow \quad m_e v = eBR$$

so the maximum kinetic energy is seen to be:

$$K_{\max} = \frac{1}{2} m_e v^2 = \frac{\left(m_e v\right)^2}{2m_e} = \frac{e^2 B^2 R^2}{2m_e}$$

$$= \frac{\left(1.602 \times 10^{-19}\ \text{C}\right)^2 \left(2.00 \times 10^{-5}\ \text{T}\right)^2 (0.200\ \text{m})^2}{2\left(9.11 \times 10^{-31}\ \text{kg}\right)}$$

$$= 2.25 \times 10^{-19}\ \text{J} = 1.40\ \text{eV}$$

From the photoelectric equation,

$$K_{\max} = hf - \phi = \frac{hc}{\lambda} - \phi$$

Thus, the work function is

$$\phi = \frac{hc}{\lambda} - K_{\max} = \frac{1\ 240\ \text{eV} \cdot \text{nm}}{450\ \text{nm}} - 1.40\ \text{eV} = \boxed{1.36\ \text{eV}}$$

P40.62 From the circular path the electrons follow in the magnetic field, the magnetic force is centripetal,

$$F = ma: \qquad evB = \frac{m_e v^2}{R} \quad \rightarrow \quad m_e v = eBR$$

so maximum kinetic energy is seen to be:

$$K_{\max} = \frac{1}{2} m_e v^2 = \frac{\left(m_e v\right)^2}{2m_e} = \frac{e^2 B^2 R^2}{2m_e}$$

From the photoelectric equation,

$$K_{\max} = hf - \phi = \frac{hc}{\lambda} - \phi$$

Thus, the work function is

$$\phi = \frac{hc}{\lambda} - K_{\max} = \boxed{\frac{hc}{\lambda} - \frac{e^2 B^2 R^2}{2m_e}}$$

P40.63 The condition on electric power delivered to the filament is

$$P = I\,\Delta V = \frac{(\Delta V)^2}{R} = \frac{(\Delta V)^2 A}{\rho \ell} = \frac{(\Delta V)^2 \pi r^2}{\rho \ell}, \quad \text{so} \quad \ell = \frac{(\Delta V)^2 \pi r^2}{\rho P}.$$

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Here $P = 75.0\text{ W},\ \rho = 7.13\times10^{-7}\,\Omega\cdot\text{m},$ and $\Delta V = 120\text{ V}$. As the filament radiates in steady state, it must emit all of this power through its lateral surface area $P = \sigma e A T^4 = \sigma e 2\pi r \ell T^4$.

(a) We combine the conditions by substitution:

$$P = \sigma e 2\pi r\left[\frac{(\Delta V)^2\pi r^2}{\rho P}\right]T^4$$

$$r^3 = \frac{\rho P^2}{2\sigma e(\Delta V)^2\pi^2 T^4}$$

$$= \frac{\left(7.13\times10^{-7}\,\Omega\cdot\text{m}\right)(75.0\text{ W})^2}{2\left(5.67\times10^{-8}\text{ W/m}^2\text{K}^4\right)(0.450)(120\text{ V})^2\pi^2(2\ 900\text{ K})^4}$$

$$r = \boxed{1.98\times10^{-5}\text{ m}}$$

$$r = \left(\frac{P\rho\ell}{\pi(\Delta V)^2}\right)^{1/2} = \left[\frac{(75.0\text{ W})\ \left(7.13\times10^{-7}\,\Omega\cdot\text{m}\right)\ (0.333\text{ m})}{\pi(120\text{ V})^2}\right]^{1/2}$$

$$= \boxed{1.98\times10^{-5}\text{ m}}$$

(b) $$\ell = \frac{(\Delta V)^2\pi r^2}{\rho P} = \frac{(120\text{ V})^2\pi r^2}{\left(7.13\times10^{-7}\,\Omega\cdot\text{m}\right)(75.0\text{ W})} = \boxed{0.333\text{ m}}$$

P40.64 We first isolate the terms involving ϕ in Equations 40.13 and 40.14,

$$\gamma m_e u\cos\phi = \frac{h}{\lambda_0} - \frac{h}{\lambda'}\cos\theta$$

$$\gamma m_e u\sin\phi = \frac{h}{\lambda'}\sin\theta$$

We then square and add to eliminate ϕ:

$$\left(\gamma m_e u\cos\phi\right)^2 + \left(\gamma m_e u\sin\phi\right)^2 = \left(\frac{h}{\lambda_0} - \frac{h}{\lambda'}\cos\theta\right)^2 + \left(\frac{h}{\lambda'}\sin\theta\right)^2$$

$$\gamma^2 m_e^2 u^2 = h^2\left[\frac{1}{\lambda_0^2} + \frac{1}{\lambda'^2} - \frac{2\cos\theta}{\lambda_0\lambda'}\right]$$

$$\frac{u^2/c^2}{\left(1-u^2/c^2\right)} = \frac{h^2}{m_e^2c^2}\left[\frac{1}{\lambda_0^2} + \frac{1}{\lambda'^2} - \frac{2\cos\theta}{\lambda_0\lambda'}\right]$$

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Defining $b = \frac{h^2}{m_e^2 c^2}\left[\frac{1}{\lambda_0^2} + \frac{1}{\lambda'^2} - \frac{2\cos\theta}{\lambda_0\lambda'}\right]$, the above equation becomes

$$\frac{u^2/c^2}{\left(1-u^2/c^2\right)} = b \qquad \rightarrow \qquad u^2/c^2 = b\left(1-u^2/c^2\right)$$

$$u^2/c^2 = \frac{b}{(1+b)}$$

Substitute into Equation 40.12 for the cutoff wavelength,

$$1+\left(\frac{h}{m_e c}\right)\left[\frac{1}{\lambda_0} - \frac{1}{\lambda'}\right] = \gamma = \left(1-\frac{b}{1+b}\right)^{-1/2} = \sqrt{1+b}$$

Squaring each side then gives

$$1+\frac{2h}{m_e c}\left[\frac{1}{\lambda_0} - \frac{1}{\lambda'}\right] + \frac{h^2}{m_e^2 c^2}\left[\frac{1}{\lambda_0} - \frac{1}{\lambda'}\right]^2 = 1+\left(\frac{h^2}{m_e^2 c^2}\right)\left[\frac{1}{\lambda_0^2} + \frac{1}{\lambda'^2} - \frac{2\cos\theta}{\lambda_0\lambda'}\right]$$

Eliminating terms,

$$\cancel{1} + \frac{2h}{m_e c}\left[\frac{1}{\lambda_0} - \frac{1}{\lambda'}\right] + \frac{h^2}{m_e^2 c^2}\left[\cancel{\frac{1}{\lambda_0^2}} - \frac{2}{\lambda_0\lambda'} + \cancel{\frac{1}{\lambda'^2}}\right] = \cancel{1} + \left(\frac{h^2}{m_e^2 c^2}\right)\left[\cancel{\frac{1}{\lambda_0^2}} + \cancel{\frac{1}{\lambda'^2}} - \frac{2\cos\theta}{\lambda_0\lambda'}\right]$$

$$\frac{\cancel{2}h}{m_e c}\left[\frac{1}{\lambda_0} - \frac{1}{\lambda'}\right] - \left(\frac{h^2}{m_e^2 c^2}\right)\left(\frac{\cancel{2}}{\lambda_0\lambda'}\right) = -\left(\frac{h^2}{m_e^2 c^2}\right)\left(\frac{\cancel{2}\cos\theta}{\lambda_0\lambda'}\right)$$

$$m_e c\left[\frac{\lambda'-\lambda_0}{\cancel{\lambda_0\lambda'}}\right] - h\left(\frac{1}{\cancel{\lambda_0\lambda'}}\right) = -h\left(\frac{\cos\theta}{\cancel{\lambda_0\lambda'}}\right)$$

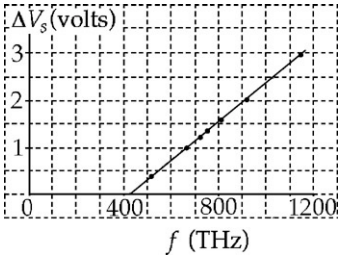
$$m_e c\left(\lambda'-\lambda_0\right) - h = -h\cos\theta$$

Rearranging this gives Equation 40.11,

$$\lambda'-\lambda_0 = \left(\frac{h}{m_e c}\right)(1-\cos\theta)$$

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P40.65 We use $\Delta V_S = \left(\frac{h}{e}\right) f - \frac{\phi}{e}$.



ANS. FIG. P40.65

From two points on the graph in ANS. FIG. P40.65,

$$0 = \left(\frac{h}{e}\right)\left(4.1 \times 10^{14}\ \text{Hz}\right) - \frac{\phi}{e}$$

and

$$3.3\ \text{V} = \left(\frac{h}{e}\right)\left(12 \times 10^{14}\ \text{Hz}\right) - \frac{\phi}{e}$$

Combining these two expressions we find:

(a) $\phi = \boxed{1.7\ \text{eV}}$

(b) $\frac{h}{e} = \boxed{4.2 \times 10^{-15}\ \text{V} \cdot \text{s}}$

(c) At the cutoff wavelength, $\frac{hc}{\lambda_c} = \phi = \left(\frac{h}{e}\right)\frac{ec}{\lambda_c}$, or

$$\lambda_c = \left(4.2 \times 10^{-15}\ \text{V} \cdot \text{s}\right)\left(1.60 \times 10^{-19}\ \text{C}\right) \times \frac{\left(3.00 \times 10^{8}\ \text{m/s}\right)}{(1.7\ \text{eV})\left(1.60 \times 10^{-19}\ \text{J/eV}\right)}$$

$$= \boxed{7.3 \times 10^{2}\ \text{nm}}$$

P40.66 Equation 40.11 states $\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta) = \lambda' - \lambda_0$ for the scattered photon. The initial energy of a photon is $E_0 = hc/\lambda_0$. Its energy after scattering is

$$E' = \frac{hc}{\lambda'} = \frac{hc}{\lambda_0 + \Delta\lambda} = hc\left[\lambda_0 + \frac{h}{m_e c}(1 - \cos\theta)\right]^{-1}$$

$$E' = \frac{hc}{\lambda_0}\left[1 + \frac{hc}{m_e c^2 \lambda_0}(1 - \cos\theta)\right]^{-1}$$

$$E' = \frac{hc}{\lambda_0}\left[1 + \frac{hc}{m_e c^2 \lambda_0}(1 - \cos\theta)\right]^{-1} = E_0\left[1 + \frac{E_0}{m_e c^2}(1 - \cos\theta)\right]^{-1}$$

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P40.67 (a) We use energy conservation in the daredevil-Earth system to find the speed of the daredevil just before he makes a splash:

$$mgy_i = \frac{1}{2}mu_f^2$$

gives

$$u_f = \sqrt{2gy_i} = \sqrt{2\left(9.80\ \text{m/s}^2\right)(50.0\ \text{m})} = 31.3\ \text{m/s}$$

The de Broglie wavelength is then

$$\lambda = \frac{h}{mu} = \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{(75.0\ \text{kg})(31.3\ \text{m/s})} = \boxed{2.82\times10^{-37}\ \text{m}}$$

This is too small to be observable.

(b) Equation 40.26 gives us the energy-lifetime version of the uncertainty principle:

$$\Delta E\Delta t \geq \frac{\hbar}{2}$$

substituting numerical values,

$$\Delta E \geq \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{4\pi\left(5.00\times10^{-3}\ \text{s}\right)} = \boxed{1.05\times10^{-32}\ \text{J}}$$

(c) We find the percent error from

$$\frac{\Delta E}{E} = \frac{1.05\times10^{-32}\ \text{J}}{(75.0\ \text{kg})\left(9.80\ \text{m/s}^2\right)(50.0\ \text{m})} = \boxed{2.87\times10^{-35}\%}$$

P40.68 The definition of the Compton wavelength is $\lambda_C = h/m_e c$. The de Broglie wavelength is $\lambda = h/p$. We take the ratio of the Compton wavelength to the de Broglie wavelength, and square it:

$$\left(\frac{\lambda_C}{\lambda}\right)^2 = \frac{p^2}{\left(m_e c\right)^2}$$

From Equation 39.27, the momentum for a slowly-moving or rapidly-moving object is described by

$$p^2 = \frac{E^2 - m_e{}^2c^4}{c^2}$$

Substituting and simplifying,

$$\left(\frac{\lambda_C}{\lambda}\right)^2 = \frac{\left(E^2 - m_e^2c^4\right)}{\left(m_e c^2\right)^2} = \left(\frac{E}{m_e c^2}\right)^2 - 1$$

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and $$\frac{\lambda_C}{\lambda} = \sqrt{\left(\frac{E}{m_e c^2}\right)^2 - 1}$$

P40.69 (a) We find the energy of one photon:

$$hf = K_{max} + \phi$$
$$= \frac{1}{2}\left(9.11\times10^{-31}\text{ kg}\right)\left(420\times10^{3}\text{ m/s}\right)^2 + (3.44\text{ eV})\left(\frac{1.6\times10^{-19}\text{ J}}{1\text{ eV}}\right) = 6.31\times10^{-19}\text{ J}$$

The number intensity of photon bombardment is

$$\frac{I}{hf} = \frac{550\text{ J/s}\cdot\text{m}^2}{6.31\times10^{-19}\text{ J/photon}}\left(\frac{1\text{ m}^2}{10^4\text{ cm}^2}\right)\left(\frac{1\text{ electron emitted}}{1\text{ photon absorbed}}\right)$$
$$= \boxed{8.72\times10^{16}\ \frac{\text{electrons}}{\text{s}\cdot\text{cm}^2}}$$

(b) The density of the current the imagined electrons comprise is

$$J = \left(8.72\times10^{16}\ \frac{\text{electrons}}{\text{s}\cdot\text{cm}^2}\right)\left(1.60\times10^{-19}\ \frac{\text{C}}{\text{electron}}\right)$$
$$= 0.014\,0\ \frac{\text{C}}{\text{s}\cdot\text{cm}^2} = \boxed{14.0\text{ mA/cm}^2}$$

(c) Many photons are likely reflected or give their energy to the metal as internal energy, so the actual current is probably a small fraction of 14.0 mA.

P40.70 From the uncertainty principle,

$$\Delta E\,\Delta t \geq \frac{\hbar}{2} \quad \rightarrow \quad \Delta\left(mc^2\right)\Delta t = \frac{\hbar}{2}$$

Therefore,

$$\frac{\Delta m}{m} = \frac{h}{4\pi c^2(\Delta t)m} = \frac{h}{4\pi(\Delta t)E_R}$$
$$= \frac{6.626\times10^{-34}\text{ J}\cdot\text{s}}{4\pi\left(8.70\times10^{-17}\text{ s}\right)(135\text{ MeV})}\left(\frac{1\text{ MeV}}{1.60\times10^{-13}\text{ J}}\right)$$
$$= \boxed{2.81\times10^{-8}}$$

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P40.71 (a) To find the de Broglie wavelength of the neutron, we first determine its momentum,

$$p = mu = \sqrt{2mE}$$
$$= \sqrt{2\left(1.67\times10^{-27}\ \text{kg}\right)\left(0.040\,0\ \text{eV}\right)\left(1.60\times10^{-19}\ \text{J/eV}\right)}$$
$$= 4.62\times10^{-24}\ \text{kg}\cdot\text{m/s}$$

Then,

$$\lambda = \frac{h}{mu} = \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{4.62\times10^{-24}\ \text{kg}\cdot\text{m/s}} = 1.43\times10^{-10}\ \text{m} = \boxed{0.143\ \text{nm}}$$

(b) This is of the same order of magnitude as the spacing between atoms in a crystal.

(c) Because the wavelength is about the same as the spacing, diffraction effects should occur.

A diffraction pattern with maxima and minima at the same angles can be produced with x-rays, with neutrons, and with electrons of much higher kinetic energy, by using incident quantum particles with the same wavelength.

Challenge Problems

***P40.72** (a) At the top of the ladder, the woman holds a pellet inside a small region Δx_i. Thus, the uncertainty principle requires her to release it with typical horizontal momentum $\Delta p_x = m\Delta v_x = \dfrac{\hbar}{2\Delta x_i}$. It falls to the floor in a travel time given by $H = 0 + \dfrac{1}{2}gt^2$ as $t = \sqrt{\dfrac{2H}{g}}$, so the total width of the impact points is

$$\Delta x_f = \Delta x_i + (\Delta v_x)t = \Delta x_i + \left(\frac{\hbar}{2m\Delta x_i}\right)\sqrt{\frac{2H}{g}} = \Delta x_i + \frac{A}{\Delta x_i}$$

where $A = \dfrac{\hbar}{2m}\sqrt{\dfrac{2H}{g}}$.

To minimize Δx_f, we require $\dfrac{d(\Delta x_f)}{d(\Delta x_i)} = 0$ or $1 - \dfrac{A}{\Delta x_i^2} = 0$,

so $\Delta x_i = \sqrt{A}$.

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The minimum width of the impact points is

$$\left(\Delta x_f\right)_{\min} = \left(\Delta x_i + \frac{A}{\Delta x_i}\right)\Bigg|_{\Delta x_i = \sqrt{A}} = 2\sqrt{A} = \boxed{\sqrt{\frac{2\hbar}{m}}\left(\frac{2H}{g}\right)^{1/4}}$$

(b) $$\left(\Delta x_f\right)_{\min} = \left[\frac{2\left(1.054\ 6\times 10^{-34}\ \text{J}\cdot\text{s}\right)}{5.00\times 10^{-4}\ \text{kg}}\right]^{1/2}\left[\frac{2(2.00\ \text{m})}{9.80\ \text{m/s}^2}\right]^{1/4}$$

$$= \boxed{5.19\times 10^{-16}\ \text{m}}$$

P40.73 (a) The Doppler shift increases the apparent frequency of the incident light.

(b) If $v = 0.280c$,

$$f' = f\sqrt{\frac{1+v/c}{1-v/c}} = \left(7.00\times 10^{14}\ \text{Hz}\right)\sqrt{\frac{1.28}{0.720}} = 9.33\times 10^{14}\ \text{Hz}$$

Therefore,

$$\phi = hf'$$

$$= \left(6.626\times 10^{-34}\ \text{J}\cdot\text{s}\right)\left(9.33\times 10^{14}\ \text{Hz}\right)\left(\frac{1\ \text{eV}}{1.602\times 10^{-19}\ \text{J}}\right)$$

$$= \boxed{3.86\ \text{eV}}$$

(c) At $v = 0.900c$,

$$f' = f\sqrt{\frac{1+v/c}{1-v/c}} = \left(7.00\times 10^{14}\ \text{Hz}\right)\sqrt{\frac{1.900}{0.100}} = 3.05\times 10^{15}\ \text{Hz}$$

and

$$K_{\max} = hf' - \phi$$

$$= \left[\left(6.626\times 10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.05\times 10^{15}\ \text{Hz}\right)\left(\frac{1.00\ \text{eV}}{1.602\times 10^{-19}\ \text{J}}\right)\right] - 3.86\ \text{eV}$$

$$= \boxed{8.76\ \text{eV}}$$

P40.74 We show that if all of the energy of a photon is transmitted to an electron, momentum will not be conserved. In general, a photon of energy $E_0 = hc/\lambda_0$ scatters off an electron at rest, resulting in the photon having energy $E' = hc/\lambda'$ and the electron having kinetic energy K_e. Energy conservation requires $E_0 = E' + K_e$, or

$$\frac{hc}{\lambda_0} = \frac{hc}{\lambda'} + m_e c^2\left(\gamma - 1\right)$$

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If the photon is absorbed, then $E' = hc/\lambda' = 0$, and the above equation becomes

$$\frac{hc}{\lambda_0} = m_e c^2 (\gamma - 1) \quad \textbf{[1]}$$

Because the photon is absorbed, momentum conservation requires the momentum of the electron be in the same direction as the momentum of the original photon:

$$p_0 = \frac{E}{c} = \frac{h}{\lambda_0} = \gamma m_e u \quad \textbf{[2]}$$

From [1], we find that

$$\gamma = \frac{h}{\lambda_0 m_e c} + 1 \quad \textbf{[3]}$$

and

$$u = c\sqrt{1 - \left(\frac{\lambda_0 m_e c}{h + \lambda_0 m_e c}\right)^2} \quad \textbf{[4]}$$

Substituting [3] and [4] into [2] reveals the inconsistency:

$$\frac{h}{\lambda_0} = \left(1 + \frac{h}{\lambda_0 m_e c}\right) m_e c \sqrt{1 - \left(\frac{\lambda_0 m_e c}{h + \lambda_0 m_e c}\right)^2}$$

$$= \frac{\lambda_0 m_e c + h}{\lambda_0} \sqrt{\frac{h(h + 2\lambda_0 m_e c)}{(h + \lambda_0 m_e c)^2}} = \frac{h}{\lambda_0}\sqrt{\frac{h + 2\lambda_0 m_e c}{h}}$$

This is impossible, so all of the energy of a photon cannot be transmitted to an electron.

P40.75 (a) Starting with Planck's law,

$$I(\lambda, T) = \frac{2\pi hc^2}{\lambda^5 \left[e^{hc/\lambda k_B T} - 1\right]}$$

the total power radiated per unit area

$$\int_0^\infty I(\lambda, T)\, d\lambda = \int_0^\infty \frac{2\pi hc^2}{\lambda^5 \left[e^{hc/\lambda k_B T} - 1\right]} d\lambda$$

Change variables by letting $x = \dfrac{hc}{\lambda k_B T}$, so $dx = -\dfrac{hc}{k_B T \lambda^2} d\lambda$.

Note that as λ varies from $0 \to \infty$, x varies from $\infty \to 0$.

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Then,

$$\int_0^\infty I(\lambda,\ T)\,d\lambda = -\frac{2\pi k_B^4 T^4}{h^3 c^2}\int_\infty^0 \frac{x^3}{\left(e^x-1\right)}dx = \frac{2\pi k_B^4 T^4}{h^3 c^2}\left(\frac{\pi^4}{15}\right)$$

Therefore,

$$\boxed{\int_0^\infty I(\lambda,\ T)\,d\lambda = \left(\frac{2\pi^5 k_B^4}{15h^3 c^2}\right)T^4 = \sigma T^4}$$

(b) From part (a),

$$\sigma = \frac{2\pi^5 k_B^4}{15h^3 c^2} = \frac{2\pi^5\left(1.38\times10^{-23}\ \text{J/K}\right)^4}{15\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)^3\left(3.00\times10^{8}\ \text{m/s}\right)^2}$$

$$\sigma = \boxed{5.67\times10^{-8}\ \text{W/m}^2\cdot\text{K}^4}$$

P40.76 (a) Planck's law states

$$I(\lambda,\ T) = \frac{2\pi hc^2}{\lambda^5\left[e^{hc/\lambda k_B T}-1\right]} = 2\pi hc^2\lambda^{-5}\left[e^{hc/\lambda k_B T}-1\right]^{-1}.$$

To find the wavelength at which this distribution has a maximum, compute

$$\frac{dI}{d\lambda} = 2\pi hc^2\left\{-5\lambda^{-6}\left[e^{hc/\lambda k_B T}-1\right]^{-1} - \lambda^{-5}\left[e^{hc/\lambda k_B T}-1\right]^{-2}e^{hc/\lambda k_B T}\left(-\frac{hc}{\lambda^2 k_B T}\right)\right\} = 0$$

$$\frac{dI}{d\lambda} = \frac{2\pi hc^2}{\lambda^6\left[e^{hc/\lambda k_B T}-1\right]}\left\{-5+\frac{hc}{\lambda k_B T}\frac{e^{hc/\lambda k_B T}}{\left[e^{hc/\lambda k_B T}-1\right]}\right\} = 0$$

Letting $x = \dfrac{hc}{\lambda k_B T}$, the condition for a maximum becomes

$\dfrac{xe^x}{e^x-1} = 5$. We zero in on the solution to this transcendental equation by iterations as shown in the table on the following page.

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x	$xe^x/(e^x-1)$
4.000 00	4.074 629 4
4.500 00	4.550 552 1
5.000 00	5.033 918 3
4.900 00	4.936 762 0
4.950 00	4.985 313 0
4.975 00	5.009 609 0
4.963 00	4.997 945 2
4.969 00	5.003 776 7
4.966 00	5.000 860 9

x	$xe^x/(e^x-1)$
4.964 50	4.999 403 0
4.965 50	5.000 374 9
4.965 00	4.999 889 0
4.965 25	5.000 132 0
4.965 13	5.000 015 3
4.965 07	4.999 957 0
4.965 10	4.999 986 2
4.965 115	5.000 000 8

The solution is found to be

$$x = \frac{hc}{\lambda_{\max} k_B T} = 4.965\,115 \qquad \text{and} \qquad \lambda_{\max} T = \frac{hc}{4.965\,115 k_B}$$

(b) Thus,

$$\lambda_{\max} T = \frac{(6.626\,075 \times 10^{-34}\ \text{J}\cdot\text{s})(2.997\,925 \times 10^{8}\ \text{m/s})}{4.965\,115(1.380\,658 \times 10^{-23}\ \text{J/K})}$$

$$= \boxed{2.897\,755 \times 10^{-3}\ \text{m}\cdot\text{K}}$$

This result agrees with Wien's experimental value of $\lambda_{\max} T = 2.898 \times 10^{-3}\ \text{m}\cdot\text{K}$ for this constant.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P40.2 (a) 999 nm; (b) The wavelength emitted at the greatest intensity is in the infrared (greater than 700 nm), and according to the graph in Active Figure 40.3, much more energy is radiated at wavelengths longer than λ_{max} than at shorter wavelengths.

P40.4 (a) 5 200 K; (b) This is not blackbody radiation.

P40.6 i: (a) 2.57 eV, (b) 1.28×10^{-5} eV, (c) 1.91×10^{-7} eV; ii: (a) 484 nm, (b) 9.68 cm, (c) 6.52 m; iii: (a) visible light (blue), (b) radio wave, (c) radio wave

P40.8 2.27×10^{30} photon/s

P40.10 (a) 5.78×10^{3} K; (b) 501 nm

P40.12 (a) 7.09×10^{4} W; (b) 580 nm; (c) 7.99×10^{10} W/m; (d–i) See table in P40.12; (j) ≈ 19 kW

P40.14 See P40.14 for full explanation.

P40.16 (a) 4.20 mm; (b) 1.05×10^{19} photons; (c) 8.82×10^{16} mm^{-3}

P40.18 (a) 288 nm; (b) 1.04×10^{15} Hz; (c) 1.19 eV

P40.20 (a) The energy of a photon with wavelength 400 nm is calculated to be 3.11 eV. Now compare this energy with the given work functions. Of these metals, only lithium shows the photoelectric effect because its work function is less than the energy of the photon; (b) 0.808 eV

P40.22 (a) 148 days; (b) The result for part (a) does not agree at all with the experimental observations.

P40.24 (a) 8.27 eV; (b) The photon energy is larger than the work function; (c) 1.92 eV; (d) 1.92 V

P40.26 4.85×10^{-12} m

P40.28 (a and b) See P40.28 for full answer; (c) 180°. We could answer like this: The photon imparts the greatest momentum to the originally stationary electron in a head-on collision. Here the photon recoils straight back, and the electron has maximum kinetic energy.

P40.30 (a) 2.89 pm; (b) $\theta = 101°$

P40.32 (a) $\theta = \cos^{-1}\left(\dfrac{m_e c^2 + E_0}{2m_e c^2 + E_0}\right)$; (b) $E' = \dfrac{E_0\left(2m_e c^2 + E_0\right)}{2\left(m_e c^2 + E_0\right)}$, $p' = \dfrac{E_0\left(2m_e c^2 + E_0\right)}{2c\left(m_e c^2 + E_0\right)}$;

(c) $K_e = \dfrac{E_0^2}{2\left(m_e c^2 + E_0\right)}$, $p_e = \dfrac{E_0\left(2m_e c^2 + E_0\right)}{2c\left(m_e c^2 + E_0\right)}$

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P40.34 (a) It is because Compton's equation and the conservation of vector momentum give three independent equations in the unknowns λ', λ_0, and u; (b) 3.82 pm

P40.36 (a) 0.667; (b) 0.001 09

P40.38 (a) 14.0 kV/m; (b) $46.8\ \mu$T; (c) 4.19 nN; (d) 10.2 g

P40.40 (a) 0.709 nm; (b) 413 nm

P40.42 3.97×10^{-13} m

P40.44 (a) ~10^{8} eV; (b) ~ -10^{6} eV; (c) The electron could not be confined to the nucleus.

P40.46 The speed with which the student passes through the door is an extremely low velocity. It is impossible for the student to walk this slowly. At this speed, if the thickness of the wall in which the door is built is 15 cm, the time interval required for the student to pass through the door is 1.4×10^{33} s, which is 10^{15} times the age of the Universe.

P40.48 (a) See P40.48(a) for full explanation; (b) They will always have a different frequency from photons of the same wavelength.

P40.50 See P40.50 for the full explanation.

P40.52 2.27×10^{-12} A

P40.54 (a) 0.250 m/s; (b) 2.25 m

P40.56 7.03×10^{-32} m/s

P40.58 For the rifles fired at targets at reasonable distances away, a spread of 1.00 cm *due to the uncertainty principle* would be impossible.

P40.60 (a) See graph in ANS. FIG. P40.60 (b) 6.4×10^{-34} J·s; (c) 1.4 eV

P40.62 $\dfrac{hc}{\lambda} - \dfrac{e^2B^2R^2}{2m_e}$

P40.64 See P40.64 for full explanation.

P40.66 See P40.66 for full explanation.

P40.68 See P40.68 for full explanation.

P40.70 2.81×10^{-8}

P40.72 (a) See P40.72(a) for full explanation; (b) 5.19×10^{-16} m

P40.74 See P40.74 for full explanation.

P40.76 (a) See P40.76 for full explanation; (b) $2.897\,755\times10^{-3}$ m·K

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